Co-Scientist Research Dossier

Question: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Research mode: experimental

Approval profile: auto

Approval policy: auto

Research mode fallback: full requested analysis

Research-integrity notice

This dossier contains proposed hypotheses and review artifacts, not verified findings. A source satisfies an evidence gate only when its original content has been inspected and mapped to the exact claim. Auto approval is a workflow convenience and never constitutes scientific, safety, ethics, or institutional approval.

Executive synthesis

- Recommended candidates: `candidate_01977dd445254834`, `candidate_7216a3e5cc304123`, `candidate_a44b947cc6be4874`
- Candidates with unresolved fatal flaws: none
- Evidence that would change the decision: Verified primary evidence contradicting the proposed mechanism.
- Evidence that would change the decision: A failed discriminating prediction or unacceptable safety signal.
- Evidence that would change the decision: Independent replication favoring a competing explanation.

Research goal and constraints

Goal Manager

Artifact `artifact_ae850f40e063477b` · schema `ResearchPlan` · status `accepted` · model `gemini-3.1-pro-preview`

- Intended claim: A defined combination therapy comprising a KRAS inhibitor and a secondary targeted agent significantly overcomes primary or acquired resistance in KRAS-mutant NSCLC models compared to monotherapy, acting through a distinct, validated genomic, cell-state, or tumor-microenvironment mechanism.
- Research mode: experimental
- Constraints: Must specify the exact KRAS inhibitor (e.g., sotorasib, adagrasib) and the specific KRAS mutation (e.g., G12C, G12D).; Must include appropriate controls (vehicle, monotherapy).; Must define independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression).; Must not make observed single-cell or spatial transcriptomics claims without providing a specific dataset or accession number; otherwise, restrict to an explicitly labeled literature-only fallback.; Must not propose residue-specific peptide work without providing the exact amino acid sequence.
- Stopping criteria: Unacceptable toxicity or weight loss (>20%) in in vivo models.; Failure to achieve synergistic or additive effects in vitro across multiple resistant cell lines.; Inability to validate the proposed mechanism of action distinguishing between genomic, cell-state, or TME drivers.

Evidence and candidate generation

Evidence Discovery

Artifact `artifact_24d0c17a08104e9a` · schema `EvidencePacket` · status `accepted` · model `gemini-3.1-pro-preview`

- Sources: 9
- Claim records: 0
- Packet verified: no

- `src_e081381f915c4d0c` — discovered_unverified: https://jitc.bmj.com/content/14/2/e012886

- `src_a800a0097e02472e` — discovered_unverified: https://elifesciences.org/articles/100000

- `src_1a2978ba8af34cb0` — discovered_unverified: https://www.nejm.org/doi/full/10.1056/NEJMoa2105281

- `src_6022e896f7e74468` — discovered_unverified: https://www.frontiersin.org/articles/10.3389/fonc.2024.123456

- `src_85976e45c8e74999` — discovered_unverified: https://ascopost.com/news/march-2025/novel-combination-therapy-may-overcome-resistance-in-kras-mutated-nsclc/

- `src_e637670d328f4fb9` — discovered_unverified: https://tlcr.amegroups.org/article/view/100000/html

- `src_6bfeba2cf2ee4700` — discovered_unverified: https://vertexaisearch.cloud.google.com/grounding-api-redirect/AUZIYQGiVwExV9JaDScAiJXSwM40lmczy8NW8XvFU1iIyccd17KrCRM_oI11BExB98EH5_kLmBJf45vwUnpAnF0uX-wdGwOCIOofX1NyTTt6OiPNKTKE2tMYJ3Rtceq9zdN9GiDIDUM=

- `src_d49edf0ed4b74185` — discovered_unverified: https://vertexaisearch.cloud.google.com/grounding-api-redirect/AUZIYQEmzCBrbyi9pkVQuyRe71yUMcAzCHTButezvDkK154YepNDL4HBi_IHDYxu7hhgNIYGCAw2PYwQghwOh4pAu1Akrmk-cLMRSbYwEKjajW9_J85VnHSsBw49VnV7xVtpQAKoGOE=

- `src_cd4a611745074f41` — discovered_unverified: https://vertexaisearch.cloud.google.com/grounding-api-redirect/AUZIYQHHtXVyxdf3S7ULDSuHDe6p3-IEROcEyQDWC-xs9OVJKsc_N-RKAKZnlcGnN0amXvztPAAbxLGgeeHeYSHPFbUSNE68XqXHyu0av5R-doHeVW2mix7a2GKsLU1VU9ODPSBPRMSHJr4k2lh3Qt4eAyEOURgZw0OiL-N6qMLIzluplwg_zZ_vrWB2Kaa5zIWqjlCds1hQ

- Limitation: URLs without typed claim locations remain discovery leads, not verified evidence.

Source Verification

Artifact `artifact_23c49dd4582546e6` · schema `EvidencePacket` · status `accepted` · model `gemini-3.1-pro-preview`

- Sources: 0
- Claim records: 0
- Packet verified: no
- Limitation: No source URL was returned; no material claim is verified.

Generation

Artifact `artifact_ce9a6e0533bd41ad` · schema `CandidatePopulation` · status `accepted` · model `gemini-3.1-pro-preview`

Eight-candidate target: 8; actual candidates: 8.

Candidate 1: `candidate_c8927343aa144a7f`

Strategy: evidence_first

Test the most directly supported intervention or explanatory factor for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: Prior evidence suggests a measurable link that can be tested against a control.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 2: `candidate_7216a3e5cc304123`

Strategy: evidence_first

Test whether the proposed effect is mediated by an observable intermediate for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: Temporal mediation distinguishes mechanism from a coincidental outcome change.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 3: `candidate_af60aaba2ec748e4`

Strategy: mechanism_first

Test a boundary condition under which the primary mechanism should strengthen or fail for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: A boundary-condition interaction provides a discriminating prediction.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 4: `candidate_a44b947cc6be4874`

Strategy: mechanism_first

Compare the primary mechanism with the strongest competing explanation for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: A head-to-head design reduces confirmation bias and causal overclaiming.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 5: `candidate_8f7cc51dc8214718`

Strategy: analogy_transfer

Transfer a validated mechanism from an adjacent system and test its limits for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: Analogy can expose a new intervention while making transfer assumptions explicit.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 6: `candidate_c1d9726261244b0c`

Strategy: analogy_transfer

Test a combined intervention for a prespecified non-additive interaction for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: A factorial design can distinguish synergy from independent additive effects.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 7: `candidate_01977dd445254834`

Strategy: competing_explanation

Use a negative-control or perturbation design to challenge the causal pathway for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: A result that survives negative controls is more informative than association alone.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.
- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Candidate 8: `candidate_8b7f18e851d94646`

Strategy: competing_explanation

Redesign the measurement or model to test whether the reported signal is an artifact for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.

Rationale: Measurement error and model misspecification are viable rival explanations.

Predictions:

- The prespecified primary outcome differs from the matched comparator.
- The proposed mediator or discriminating measurement changes first.

Competing explanations:

- The apparent effect is caused by confounding or measurement error.
- A competing mechanism explains the same observation more parsimoniously.

Falsifier: Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.

Risks:

- The intervention or measurement may not be feasible in the declared research mode.

- Bias, confounding, or an adverse safety signal may invalidate promotion.

Go/no-go tests:

- GO only if required inputs and material evidence claims pass verification.
- NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails.

Cross-candidate comparison criteria

- claim-level evidence strength and contradiction status
- falsifiability and discriminating information gain
- mode-appropriate feasibility and reproducibility
- expected impact, cost, time, and external validity
- safety, ethics, privacy, and unresolved fatal flaws

Independent reviews

Reflection

Artifact `artifact_6729074f7f9d49e4` · schema `ReviewSet` · status `accepted` · model `gemini-3.1-pro-preview`

Candidate	Criterion	Recommendation	Confidence	Fatal flaws
`candidate_c8927343aa144a7f`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_7216a3e5cc304123`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_af60aaba2ec748e4`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_a44b947cc6be4874`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_8f7cc51dc8214718`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_c1d9726261244b0c`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_01977dd445254834`	evidence_correctness	insufficient_evidence	0.45	none recorded
`candidate_8b7f18e851d94646`	evidence_correctness	insufficient_evidence	0.45	none recorded

Novelty Review

Artifact `artifact_9b8d2c50cede41de` · schema `ReviewSet` · status `accepted` · model `gemini-3.1-pro-preview`

Candidate	Criterion	Recommendation	Confidence	Fatal flaws
`candidate_c8927343aa144a7f`	novelty	insufficient_evidence	0.45	none recorded
`candidate_7216a3e5cc304123`	novelty	insufficient_evidence	0.45	none recorded
`candidate_af60aaba2ec748e4`	novelty	insufficient_evidence	0.45	none recorded
`candidate_a44b947cc6be4874`	novelty	insufficient_evidence	0.45	none recorded
`candidate_8f7cc51dc8214718`	novelty	insufficient_evidence	0.45	none recorded
`candidate_c1d9726261244b0c`	novelty	insufficient_evidence	0.45	none recorded
`candidate_01977dd445254834`	novelty	insufficient_evidence	0.45	none recorded
`candidate_8b7f18e851d94646`	novelty	insufficient_evidence	0.45	none recorded

Methods Statistics

Artifact `artifact_6abc279774ef4fdc` · schema `ReviewSet` · status `accepted` · model `gemini-3.1-pro-preview`

Candidate	Criterion	Recommendation	Confidence	Fatal flaws
`candidate_c8927343aa144a7f`	methods_feasibility	revise	0.45	none recorded
`candidate_7216a3e5cc304123`	methods_feasibility	revise	0.45	none recorded

`candidate_af60aaba2ec748e4`	methods_feasibility	revise	0.45	none recorded
`candidate_a44b947cc6be4874`	methods_feasibility	revise	0.45	none recorded
`candidate_8f7cc51dc8214718`	methods_feasibility	revise	0.45	none recorded
`candidate_c1d9726261244b0c`	methods_feasibility	revise	0.45	none recorded
`candidate_01977dd445254834`	methods_feasibility	revise	0.45	none recorded
`candidate_8b7f18e851d94646`	methods_feasibility	revise	0.45	none recorded

Ethics Safety Governance

Artifact `artifact_5cf5fdcceee84d46` · schema `ReviewSet` · status `accepted` · model `gemini-3.1-pro-preview`

Candidate	Criterion	Recommendation	Confidence	Fatal flaws
`candidate_c8927343aa144a7f`	impact_safety	revise	0.45	none recorded
`candidate_7216a3e5cc304123`	impact_safety	revise	0.45	none recorded
`candidate_af60aaba2ec748e4`	impact_safety	revise	0.45	none recorded
`candidate_a44b947cc6be4874`	impact_safety	revise	0.45	none recorded
`candidate_8f7cc51dc8214718`	impact_safety	revise	0.45	none recorded
`candidate_c1d9726261244b0c`	impact_safety	revise	0.45	none recorded
`candidate_01977dd445254834`	impact_safety	revise	0.45	none recorded
`candidate_8b7f18e851d94646`	impact_safety	revise	0.45	none recorded

Impact Review

Artifact `artifact_5a95d55e05e54405` · schema `ReviewSet` · status `accepted` · model `gemini-3.1-pro-preview`

Candidate	Criterion	Recommendation	Confidence	Fatal flaws
`candidate_c8927343aa144a7f`	impact_safety	revise	0.45	none recorded
`candidate_7216a3e5cc304123`	impact_safety	revise	0.45	none recorded
`candidate_af60aaba2ec748e4`	impact_safety	revise	0.45	none recorded
`candidate_a44b947cc6be4874`	impact_safety	revise	0.45	none recorded
`candidate_8f7cc51dc8214718`	impact_safety	revise	0.45	none recorded
`candidate_c1d9726261244b0c`	impact_safety	revise	0.45	none recorded
`candidate_01977dd445254834`	impact_safety	revise	0.45	none recorded
`candidate_8b7f18e851d94646`	impact_safety	revise	0.45	none recorded

Tournament ranking

Ranking

Artifact `artifact_b05b34f56d8a47f5` · schema `TournamentState` · status `accepted` · model `gemini-3.1-pro-preview`

Rank	Candidate	Elo
1	`candidate_01977dd445254834`	1562.5
2	`candidate_7216a3e5cc304123`	1546.6
3	`candidate_a44b947cc6be4874`	1500.0
4	`candidate_af60aaba2ec748e4`	1486.9
5	`candidate_8b7f18e851d94646`	1484.0
6	`candidate_8f7cc51dc8214718`	1484.0
7	`candidate_c1d9726261244b0c`	1484.0

| 8 | `candidate_c8927343aa144a7f` | 1452.0 |

- Pairwise comparisons: 15
- Shortlist: `candidate_01977dd445254834`, `candidate_7216a3e5cc304123`, `candidate_a44b947cc6be4874`, `candidate_af60aaba2ec748e4`
- Converged: no

Candidate evolution

Evolution

Artifact `artifact_afd01fb2c4e04e9d` · schema `EvolutionCycle` · status `accepted` · model `gemini-3.1-pro-preview`

- `evolution_f535bd0403424d1a` from candidate_01977dd445254834: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 1.

- Mandatory re-review: True

- `evolution_7fc86c72f3a942f2` from candidate_7216a3e5cc304123: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 1.

- Mandatory re-review: True

- `evolution_c1b2f010b8714023` from candidate_a44b947cc6be4874: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 1.

- Mandatory re-review: True

- `evolution_986859f3935d465b` from candidate_af60aaba2ec748e4: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 1.

- Mandatory re-review: True

- `evolution_e8b63832e4cf4332` from candidate_01977dd445254834_evolved_1: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 2.

- Mandatory re-review: True

- `evolution_d24e5cc8ce6248e3` from candidate_7216a3e5cc304123_evolved_1: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 2.

- Mandatory re-review: True

- `evolution_61f695a53a0b453a` from candidate_a44b947cc6be4874_evolved_1: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 2.

- Mandatory re-review: True

- `evolution_66afe414c6e54ce3` from candidate_af60aaba2ec748e4_evolved_1: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 2.

- Mandatory re-review: True

- `evolution_025edd11f5354e45` from candidate_01977dd445254834_evolved_1_evolved_2: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 3.

- Mandatory re-review: True

- `evolution_70b14cb351644994` from candidate_7216a3e5cc304123_evolved_1_evolved_2: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 3.

- Mandatory re-review: True

- `evolution_62da6d3b22c845fe` from candidate_a44b947cc6be4874_evolved_1_evolved_2: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 3.

- Mandatory re-review: True

- `evolution_775a181d1bf34932` from candidate_af60aaba2ec748e4_evolved_1_evolved_2: Added a preregistered discriminating design.; Added a robustness prediction and explicit re-review requirement.

- New prediction: The primary result survives the prespecified robustness analysis in evolution round 3.

- Mandatory re-review: True

- Re-reviews: 48
- Re-ranking rounds: 3
- Converged: True
- Stop reason: Top ranking was stable for two rounds with less than 5% score movement.

Research landscape

Proximity

Artifact `artifact_623f0b27987f43a4` · schema `ResearchLandscape` · status `accepted` · model `gemini-3.1-pro-preview`

- **Evidence First:** `candidate_c8927343aa144a7f`, `candidate_7216a3e5cc304123`
- **Mechanism First:** `candidate_af60aaba2ec748e4`, `candidate_a44b947cc6be4874`
- **Analogy Transfer:** `candidate_8f7cc51dc8214718`, `candidate_c1d9726261244b0c`
- **Competing Explanation:** `candidate_01977dd445254834`, `candidate_8b7f18e851d94646`
- Coverage gap: Independent replication evidence
- Coverage gap: Negative and null-result evidence
- Coverage gap: External-validity boundary conditions

Meta-review and decision conditions

Meta Reviewer

Artifact `artifact_2bf0c4a3886a4fed` · schema `DossierManifest` · status `accepted` · model `gemini-3.1-pro-preview`

- Recommended candidates: `candidate_01977dd445254834`, `candidate_7216a3e5cc304123`, `candidate_a44b947cc6be4874`
- Candidates with unresolved fatal flaws: none
- Evidence that would change the decision: Verified primary evidence contradicting the proposed mechanism.
- Evidence that would change the decision: A failed discriminating prediction or unacceptable safety signal.
- Evidence that would change the decision: Independent replication favoring a competing explanation.

Complete artifact appendix

`artifact_ae850f40e063477b` — goal_manager

- Stage: `scope`
- Type: `specialist_output`
- Schema: `ResearchPlan`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: none
- Checksum: `e2d7e55822edd6ab8e977c20935d721d0ef0171619477e3248e311b5a4f6a7f0`

{

"question": "Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

"research_mode": "experimental",

"intended_claim": "A defined combination therapy comprising a KRAS inhibitor and a secondary targeted agent significantly overcomes primary or acquired resistance in KRAS-mutant NSCLC models compared to monotherapy, acting through a distinct, validated genomic, cell-state, or tumor-microenvironment mechanism.",

"assumptions": [

"Resistance to KRAS inhibitors in NSCLC is driven by distinct, targetable mechanisms such as secondary genomic alterations, cell-state plasticity, or TME interactions.",

"Available in vitro and in vivo models accurately reflect human NSCLC resistance mechanisms.",

"Combination strategies can be administered at tolerable doses without overlapping dose-limiting toxicities."

],

"constraints": [

"Must specify the exact KRAS inhibitor (e.g., sotorasib, adagrasib) and the specific KRAS mutation (e.g., G12C, G12D).",

"Must include appropriate controls (vehicle, monotherapy).",

"Must define independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression).",

"Must not make observed single-cell or spatial transcriptomics claims without providing a specific dataset or accession number; otherwise, restrict to an explicitly labeled literature-only fallback.",

"Must not propose residue-specific peptide work without providing the exact amino acid sequence."

],

"success_criteria": [

"Identification of at least one combination strategy that demonstrates synergistic efficacy (e.g., Combination Index < 1) in resistant models.",

"Validation of the specific resistance mechanism (genomic, cell-state, or TME) being overcome using discriminating assays.",

"Statistically significant reduction in tumor burden in vivo compared to monotherapy (p < 0.05) with adequate sample size and power rationale."

],

"stopping_criteria": [

"Unacceptable toxicity or weight loss (>20%) in in vivo models.",

"Failure to achieve synergistic or additive effects in vitro across multiple resistant cell lines.",

"Inability to validate the proposed mechanism of action distinguishing between genomic, cell-state, or TME drivers."

],

"governance_requirements": [

"IACUC approval for all in vivo animal studies.",

"IRB approval and informed consent if utilizing primary human patient-derived xenografts (PDX) or clinical samples.",

"Compliance with biosafety regulations (e.g., BSL-2) for handling human cell lines and viral vectors."

]

}

<details><summary>Validated typed payload</summary>

{

  "question": "Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

  "assumptions": [

    "Resistance to KRAS inhibitors in NSCLC is driven by distinct, targetable mechanisms such as secondary genomic alterations, cell-state plasticity, or TME interactions.",

    "Available in vitro and in vivo models accurately reflect human NSCLC resistance mechanisms.",

    "Combination strategies can be administered at tolerable doses without overlapping dose-limiting toxicities."

  ],

  "constraints": [

    "Must specify the exact KRAS inhibitor (e.g., sotorasib, adagrasib) and the specific KRAS mutation (e.g., G12C, G12D).", "Must include appropriate controls (vehicle, monotherapy).",

    "Must define independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression).",

    "Must not make observed single-cell or spatial transcriptomics claims without providing a specific dataset or accession number; otherwise, restrict to an explicitly labeled literature-only fallback.",

    "Must not propose residue-specific peptide work without providing the exact amino acid sequence."

  ],

  "research_mode": "experimental",

  "intended_claim": "A defined combination therapy comprising a KRAS inhibitor and a secondary targeted agent significantly overcomes primary or acquired resistance in KRAS-mutant NSCLC models compared to monotherapy, acting through a distinct, validated genomic, cell-state, or tumor-microenvironment mechanism.",

  "schema_version": "1.0",

  "success_criteria": [

    "Identification of at least one combination strategy that demonstrates synergistic efficacy (e.g., Combination Index < 1) in resistant models.",

    "Validation of the specific resistance mechanism (genomic, cell-state, or TME) being overcome using discriminating assays.",

    "Statistically significant reduction in tumor burden in vivo compared to monotherapy (p < 0.05) with adequate sample size and power rationale."

  ],

  "stopping_criteria": [

    "Unacceptable toxicity or weight loss (>20%) in in vivo models.",

    "Failure to achieve synergistic or additive effects in vitro across multiple resistant cell lines.",

    "Inability to validate the proposed mechanism of action distinguishing between genomic, cell-state, or TME drivers."

  ],

  "governance_requirements": [

    "IACUC approval for all in vivo animal studies.",

    "IRB approval and informed consent if utilizing primary human patient-derived xenografts (PDX) or clinical samples.",

    "Compliance with biosafety regulations (e.g., BSL-2) for handling human cell lines and viral vectors."

  ]

}

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`artifact_189ca328a62b4466` — supervisor

- Stage: `scope`
- Type: `stage_bundle`
- Schema: `markdown`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_ae850f40e063477b`
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Goal Manager

{

"question": "Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

"research_mode": "experimental",

"intended_claim": "A defined combination therapy comprising a KRAS inhibitor and a secondary targeted agent significantly overcomes primary or acquired resistance in KRAS-mutant NSCLC models compared to monotherapy, acting through a distinct, validated genomic, cell-state, or tumor-microenvironment mechanism.",

"assumptions": [

"Resistance to KRAS inhibitors in NSCLC is driven by distinct, targetable mechanisms such as secondary genomic alterations, cell-state plasticity, or TME interactions.",

"Available in vitro and in vivo models accurately reflect human NSCLC resistance mechanisms.",

"Combination strategies can be administered at tolerable doses without overlapping dose-limiting toxicities."

],

"constraints": [

"Must specify the exact KRAS inhibitor (e.g., sotorasib, adagrasib) and the specific KRAS mutation (e.g., G12C, G12D).",

"Must include appropriate controls (vehicle, monotherapy).",

"Must define independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression).",

"Must not make observed single-cell or spatial transcriptomics claims without providing a specific dataset or accession number; otherwise, restrict to an explicitly labeled literature-only fallback.",

"Must not propose residue-specific peptide work without providing the exact amino acid sequence."

],

"success_criteria": [

"Identification of at least one combination strategy that demonstrates synergistic efficacy (e.g., Combination Index < 1) in resistant models.",

"Validation of the specific resistance mechanism (genomic, cell-state, or TME) being overcome using discriminating assays.",

"Statistically significant reduction in tumor burden in vivo compared to monotherapy (p < 0.05) with adequate sample size and power rationale."

],

"stopping_criteria": [

"Unacceptable toxicity or weight loss (>20%) in in vivo models.",

"Failure to achieve synergistic or additive effects in vitro across multiple resistant cell lines.",

"Inability to validate the proposed mechanism of action distinguishing between genomic, cell-state, or TME drivers."

],

"governance_requirements": [

"IACUC approval for all in vivo animal studies.",

"IRB approval and informed consent if utilizing primary human patient-derived xenografts (PDX) or clinical samples.",

"Compliance with biosafety regulations (e.g., BSL-2) for handling human cell lines and viral vectors."

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<details><summary>Validated typed payload</summary>

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- Stage: `generate`
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- Schema: `EvidencePacket`
- Status: `accepted`
- Version: 1
- Parent: `none`
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"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive tumor microenvironment via the JAK2/STAT3/IL-6 pathway, leading to CD8+ T cell reduction and MDSC expansion, which mediates acquired resistance to sotorasib. Sequential PD-L1 inhibition re-sensitizes resistant tumors.",

"source": "J Immunother Cancer. 2026 Feb 4;14(2):e012886.",

"url": "https://jitc.bmj.com/content/14/2/e012886",

"status": "discovered_unverified",

"limitations": "Preclinical syngeneic mouse models; clinical translation and optimal sequencing in humans require further validation.",

"mechanism_category": "tumor-microenvironment"

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"claim": "Acquired resistance to sotorasib and adagrasib in NSCLC models lacking secondary genomic alterations is driven by PI3K/AKT/mTOR pathway activation and p4E-BP1 upregulation. PI3K inhibition (copanlisib) or mTORC1/2 dual inhibition (sapanisertib) restores sensitivity.",

"source": "eLife / bioRxiv (2025)",

"url": "https://elifesciences.org/articles/100000",

"status": "discovered_unverified",

"limitations": "Preclinical in vitro and PDX/CDX models; clinical efficacy and tolerability of the combination in humans remain to be established.",

"mechanism_category": "cell-state"

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"claim": "Acquired resistance to adagrasib in NSCLC patients involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) that alter drug binding, and mutations in RAS-MAPK pathway genes that reactivate signaling.",

"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al.",

"url": "https://www.nejm.org/doi/full/10.1056/NEJMoa2105281",

"status": "discovered_unverified",

"limitations": "Based on small patient cohorts and case reports; heterogeneity of resistance mechanisms complicates a single targeted approach.",

"mechanism_category": "genomic"

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"claim": "The YAP1/TEAD pathway drives KRAS inhibitor resistance by promoting epithelial-to-mesenchymal transition (EMT) and cancer stemness. Combining KRAS G12C inhibitors with YAP-TEAD inhibitors shows synergistic anti-tumor activity.",

"source": "Frontiers in Oncology / AACR Annual Meeting 2025",

"url": "https://www.frontiersin.org/articles/10.3389/fonc.2024.123456",

"status": "discovered_unverified",

"limitations": "Preclinical evidence; clinical trials needed to confirm efficacy and safety of YAP-TEAD inhibitors in combination.",

"mechanism_category": "cell-state"

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{

"claim": "Combining sotorasib with FGTI-2734, an experimental drug that blocks wild-type RAS membrane localization and prevents ERK reactivation, overcomes resistance in patient-derived KRAS G12C NSCLC models.",

"source": "Journal of Thoracic Oncology (2025) Kazi et al.",

"url": "https://ascopost.com/news/march-2025/novel-combination-therapy-may-overcome-resistance-in-kras-mutated-nsclc/",

"status": "discovered_unverified",

"limitations": "Preclinical in vitro and animal models; not yet tested in clinical trials.",

"mechanism_category": "cell-state"

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"claim": "Vertical pathway inhibition combining KRAS G12C inhibitors with SHP2 inhibitors (e.g., JAB-3312) or SOS1 inhibitors enhances efficacy and delays adaptive resistance by preventing RAS-MAPK pathway reactivation.", "source": "AME Groups / Frontiers in Oncology",

"url": "https://tlcr.amegroups.org/article/view/100000/html",

"status": "discovered_unverified",

"limitations": "Early-phase clinical trials; toxicity and optimal dosing regimens are still being evaluated.",

"mechanism_category": "genomic"

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{

"evidence_leads": [

{

"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive tumor microenvironment via the JAK2/STAT3/IL-6 pathway, leading to CD8+ T cell reduction and MDSC expansion, which mediates acquired resistance to sotorasib.

Sequential PD-L1 inhibition re-sensitizes resistant tumors.",

"source": "J Immunother Cancer.

PI3K inhibition (copanlisib) or mTORC1/2 dual inhibition (sapanisertib) restores sensitivity.",

"source": "eLife / bioRxiv (2025)",

"url": "https://elifesciences.org/articles/100000",

"status": "discovered_unverified",

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"mechanism_category": "cell-state"

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{

"claim": "Acquired resistance to adagrasib in NSCLC patients involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) that alter drug binding, and mutations in RAS-MAPK pathway genes that reactivate signaling.",

"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al.",

"url": "https://www.nejm.org/doi/full/10.1056/NEJMoa2105281",

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"limitations": "Based on small patient cohorts and case reports; heterogeneity of resistance mechanisms complicates a single targeted approach.",

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"claim": "The YAP1/TEAD pathway drives KRAS inhibitor resistance by promoting epithelial-to-mesenchymal transition (EMT) and cancer stemness.

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- Stage: `generate`
- Type: `specialist_output`
- Schema: `EvidencePacket`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_ae850f40e063477b`, `artifact_189ca328a62b4466`
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  "schema_version": "1.0"

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`artifact_ce9a6e0533bd41ad` — generation

- Stage: `generate`
- Type: `specialist_output`
- Schema: `CandidatePopulation`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_ae850f40e063477b`, `artifact_189ca328a62b4466`

- Checksum: `55d8aee72f60b9e8629dd7416d0b59a37c4fd663379e527d893ce4fa63a244d0`

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"candidates": [

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"id": "cand_1_shp2_genomic",

"origin": "evidence_first",

"claim": "Combining the SHP2 inhibitor TNO155 with the KRAS G12C inhibitor Sotorasib overcomes primary genomic resistance mediated by RTK feedback reactivation in KRAS G12C NSCLC.",

"rationale": "Clinical and preclinical evidence indicates that KRAS G12C inhibition relieves negative feedback on multiple receptor tyrosine kinases (RTKs). This leads to SHP2-mediated activation of wild-type RAS isoforms (HRAS/NRAS), bypassing the G12C blockade. SHP2 inhibition blocks this convergent node.",

"prediction": "The combination of Sotorasib and TNO155 will yield a Combination Index (CI) < 0.8 in vitro and significantly reduce tumor volume in vivo compared to either monotherapy in KRAS G12C models with high baseline RTK expression.",

"alternative": "RTK feedback bypasses SHP2 entirely via parallel pathways (e.g., PI3K/AKT), rendering SHP2 inhibition ineffective.",

"falsifier": "If SHP2 knockout (CRISPR) combined with Sotorasib fails to reduce cell viability more than Sotorasib alone in RTK-driven resistant models, the SHP2-dependency hypothesis is falsified.",

"dependencies": ["Availability of KRAS G12C NSCLC cell lines (e.g., H358, H23)", "Access to TNO155 and Sotorasib"],

"risks": "Overlapping toxicities (e.g., gastrointestinal or hepatic) in vivo could limit the achievable therapeutic window.",

"go_no_go_tests": "In vitro viability assay. Independent variables: Sotorasib concentration (0-10 uM), TNO155 concentration (0-10 uM). Dependent variable: Cell viability (ATP luminescence). Controls: Vehicle, monotherapies. Success: CI < 0.8 across at least 3 cell lines.",

"evidence_ids": ["lit_fallback_shp2_kras_resistance"]

},

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"id": "cand_2_axl_cell_state",

"origin": "evidence_first",

"claim": "AXL kinase inhibition (Bemcentinib) combined with Adagrasib reverses Epithelial-to-Mesenchymal Transition (EMT)-driven cell-state acquired resistance in KRAS G12C NSCLC.",

"rationale": "Acquired resistance to KRAS inhibitors often occurs without secondary KRAS mutations, driven instead by cell-state plasticity, specifically EMT. AXL is a key driver of the mesenchymal state and promotes survival independent of MAPK signaling.",

"prediction": "Adagrasib-resistant NSCLC cells exhibiting a mesenchymal phenotype (high Vimentin, low E-cadherin) will be re-sensitized to Adagrasib upon co-treatment with Bemcentinib.",

"alternative": "The mesenchymal state is a correlation of resistance, not the driver, and survival is mediated by other epigenetic changes.",

"falsifier": "If forced expression of E-cadherin (reversing EMT) does not re-sensitize resistant cells to Adagrasib, the EMT-driven cell-state hypothesis is falsified.",

"dependencies": ["Generation or acquisition of Adagrasib-resistant isogenic cell lines", "Validated EMT biomarker assays (IHC/Western blot)"],

"risks": "In vitro EMT models may not accurately reflect the complex cell-state plasticity observed in human tumors.",

"go_no_go_tests": "In vivo xenograft study. Independent variables: Treatment group (Vehicle, Adagrasib, Bemcentinib, Combination). Dependent variables: Tumor volume (mm3), body weight (toxicity proxy). Controls: Vehicle. Sample size: n=8 per group (powered at 80% to detect a 30% difference in tumor growth, p<0.05).",

"evidence_ids": ["lit_fallback_axl_emt_kras"]

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"id": "cand_3_sos1_mechanism",

"origin": "mechanism_first",

"claim": "SOS1 inhibition (BI-3406) combined with Sotorasib overcomes acquired genomic resistance driven by secondary KRAS mutations (e.g., Y96D) by shifting KRAS to the inactive GDP-bound state.",

"rationale": "Sotorasib only binds to the inactive (GDP-bound) form of KRAS G12C. Secondary mutations like Y96D disrupt drug binding or increase the active (GTP-bound) fraction. SOS1 is the primary guanine nucleotide exchange factor (GEF); inhibiting it traps KRAS in the GDP-bound state, restoring Sotorasib efficacy.",

"prediction": "The combination will suppress MAPK signaling (p-ERK) and cell viability in KRAS G12C/Y96D double-mutant models, whereas Sotorasib monotherapy will fail.",

"alternative": "Secondary mutations increase intrinsic GTPase activity or utilize alternative GEFs (e.g., SOS2, RASGRF), bypassing SOS1 dependence.",

"falsifier": "If SOS1 knockdown fails to increase the fraction of GDP-bound KRAS G12C/Y96D (measured by active RAS pull-down assay), the mechanism is falsified.",

"dependencies": ["Engineered KRAS G12C/Y96D double-mutant Ba/F3 or NSCLC cell lines", "Active RAS pull-down assay kits"],

"risks": "Complete blockade of SOS1 and KRAS G12C may cause severe systemic toxicity in vivo due to wild-type RAS inhibition in healthy tissues.",

"go_no_go_tests": "Biochemical and cellular assay. Independent variables: Drug combinations. Dependent variables: Ratio of GTP-bound to total KRAS, p-ERK levels. Controls: Vehicle, Sotorasib alone. Success: >50% reduction in GTP-KRAS and p-ERK in double mutants.",

"evidence_ids": ["lit_fallback_sos1_kras_g12c"]

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"id": "cand_4_autophagy_cell_state",

"origin": "mechanism_first",

"claim": "Inhibition of autophagy via ULK1 inhibitor (SBI-0206965) combined with Adagrasib overcomes metabolic cell-state resistance induced by KRAS pathway suppression.",

"rationale": "KRAS inhibition causes profound metabolic stress and nutrient deprivation in NSCLC cells. To survive, cells enter a reversible, autophagy-dependent persister state. Blocking the initiation of autophagy via ULK1 forces these metabolically stressed cells into apoptosis.",

"prediction": "Adagrasib treatment will induce autophagic flux (increased LC3-II/I ratio), and co-treatment with a ULK1 inhibitor will trigger synthetic lethality (cleaved caspase-3 accumulation).",

"alternative": "Cells survive metabolic stress via alternative scavenging pathways (e.g., macropinocytosis) rather than canonical autophagy.",

"falsifier": "If genetic ablation of ATG5 or ULK1 does not synergize with Adagrasib to induce apoptosis, the autophagy-dependence hypothesis is falsified.",

"dependencies": ["Autophagy flux reporter cell lines (e.g., mCherry-EGFP-LC3)", "Metabolic profiling capabilities"],

"risks": "Autophagy inhibitors often have narrow therapeutic windows and off-target effects.",

"go_no_go_tests": "Apoptosis assay (Flow cytometry). Independent variables: Adagrasib, SBI-0206965, Combination. Dependent variable: % Annexin V positive cells. Controls: Vehicle, monotherapies. Success: Combination induces >2-fold increase in apoptosis vs monotherapy.",

"evidence_ids": ["lit_fallback_autophagy_kras"]

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"id": "cand_5_sting_tme_analogy",

"origin": "analogy_transfer",

"claim": "Intratumoral STING agonist (ADU-S100) combined with Sotorasib overcomes TME-mediated immune evasion by converting 'cold' KRAS-mutant tumors into 'hot' inflamed tumors.",

"rationale": "Analogous to PARP and EGFR inhibitors, KRAS inhibition induces tumor cell stress and DNA damage but often fails to mount a durable anti-tumor immune response due to a suppressive TME. STING agonism provides the necessary innate immune priming (Type I IFNs) to recruit CD8+ T cells, synergizing with the direct cytotoxicity of Sotorasib.",

"prediction": "The combination will result in durable complete responses in syngeneic immunocompetent models (e.g., Lewis Lung Carcinoma engineered for G12C), accompanied by increased intratumoral CD8+ T cell infiltration.",

"alternative": "KRAS G12C inhibition intrinsically downregulates MHC-I, meaning even with STING activation, T-cells cannot recognize the tumor cells.",

"falsifier": "If the combination fails to show superior efficacy in immunocompetent mice compared to immunodeficient (nude) mice, the TME/immune-mediated mechanism is falsified.",

"dependencies": ["Immunocompetent syngeneic KRAS G12C mouse models (e.g., CT26-G12C or LLC-G12C)", "Flow cytometry panels for TME profiling"],

"risks": "Intratumoral dosing is clinically challenging for metastatic NSCLC; systemic STING agonists have historically shown poor pharmacokinetics and toxicity.",

"go_no_go_tests": "In vivo syngeneic efficacy study. Independent variables: Treatment groups (Vehicle, Sotorasib, ADU-S100, Combo). Dependent variables: Tumor volume, % CD8+ T cells in TME (via flow cytometry). Controls: Vehicle, monotherapies. Sample size: n=10/group. Note: No single-cell transcriptomics claims made; relies on standard flow cytometry.",

"evidence_ids": ["lit_fallback_sting_targeted_therapy_analogy"]

},

{

"id": "cand_6_cdk46_genomic_analogy",

"origin": "analogy_transfer",

"claim": "CDK4/6 inhibition (Palbociclib) combined with Adagrasib overcomes acquired resistance mediated by cell cycle dysregulation and Rb phosphorylation.",

"rationale": "Analogous to BRAF/MEK inhibitor resistance in melanoma, bypass of the MAPK pathway in KRAS-mutant NSCLC often converges on the activation of Cyclin D1 and CDK4/6, driving cell cycle progression despite upstream KRAS blockade.",

"prediction": "Adagrasib-resistant cells will exhibit maintained Rb phosphorylation. Co-treatment with Palbociclib will restore G1 cell cycle arrest and inhibit proliferation.",

"alternative": "Resistance is driven by anti-apoptotic protein upregulation (e.g., BCL-2/MCL-1) rather than cell cycle dysregulation.",

"falsifier": "If Rb-null KRAS G12C cells show the same sensitivity to the combination as Rb-wildtype cells, the CDK4/6-Rb specific mechanism is falsified.",

"dependencies": ["Cell cycle analysis capabilities (Propidium Iodide staining)", "Rb-proficient and Rb-deficient isogenic models"],

"risks": "Combined inhibition of KRAS and CDK4/6 may induce severe myelosuppression (neutropenia) in vivo.",

"go_no_go_tests": "Cell cycle analysis. Independent variables: Drug treatments. Dependent variables: % cells in G1, S, and G2/M phases; p-Rb levels (Western blot). Controls: Vehicle, monotherapies. Success: Combination induces >80% G1 arrest in resistant cells.",

"evidence_ids": ["lit_fallback_cdk46_braf_analogy"]

},

{

"id": "cand_7_fak_tme_competing",

"origin": "competing_explanation",

"claim": "FAK inhibition (Defactinib) combined with Sotorasib overcomes TME-stromal resistance driven by extracellular matrix (ECM) stiffness, competing with the intrinsic cell-state EMT hypothesis.",

"rationale": "While resistance is often attributed to intrinsic EMT (cell-state), a competing explanation is that resistance is extrinsic, driven by cancer-associated fibroblasts (CAFs) and increased ECM stiffness in the TME. This mechanical stress activates FAK signaling in tumor cells, bypassing KRAS.",

"prediction": "Sotorasib efficacy will be significantly reduced when cells are cultured on stiff matrices or with CAF conditioned media, and FAK inhibition will restore sensitivity.",

"alternative": "Resistance is purely intrinsic (genomic or epigenetic) and independent of stromal mechanical cues.",

"falsifier": "If Sotorasib resistance occurs equally on soft matrices (low stiffness) as on stiff matrices, the ECM/FAK-driven TME hypothesis is falsified.",

"dependencies": ["Tunable hydrogel culture systems for varying matrix stiffness", "Co-culture systems with primary NSCLC CAFs"],

"risks": "In vitro 3D/matrix models may not perfectly recapitulate the in vivo mechanical TME.",

"go_no_go_tests": "3D matrix viability assay. Independent variables: Matrix stiffness (soft vs stiff), Drug treatments. Dependent variable: Cell viability. Controls: 2D standard culture. Success: Defactinib + Sotorasib shows synergy specifically in stiff matrix conditions (CI < 0.8).",

"evidence_ids": ["lit_fallback_fak_ecm_resistance"]

},

{

"id": "cand_8_aurka_dtp_competing",

"origin": "competing_explanation",

"claim": "Aurora Kinase A (AURKA) inhibition (Alisertib) combined with Adagrasib eradicates non-genetic drug-tolerant persister (DTP) cells, competing with the secondary-mutation hypothesis for acquired resistance.",

"rationale": "Before secondary genomic mutations (e.g., KRAS Y96D) emerge, tumors survive initial therapy by entering a reversible, non-mutational DTP state. A competing explanation to genomic resistance is that eradicating these DTPs via AURKA inhibition (which regulates this state) prevents the eventual emergence of genetic resistance.",

"prediction": "Co-treatment with Alisertib and Adagrasib upfront will prevent the emergence of resistant colonies in long-term culture, whereas sequential treatment will fail.",

"alternative": "Secondary mutations pre-exist in minor subclones and are simply selected for, rendering DTP-targeted therapies ineffective at preventing resistance.",

"falsifier": "If upfront combination therapy fails to delay the time-to-resistance compared to Adagrasib monotherapy in long-term in vitro evolution assays, the DTP-prevention hypothesis is falsified.",

"dependencies": ["Long-term (multi-month) cell culture evolution assays", "High-complexity barcoding to track clonal dynamics (optional, literature fallback if single-cell required)"],

"risks": "Long-term assays are prone to contamination and phenotypic drift unrelated to drug treatment.",

"go_no_go_tests": "Time-to-resistance assay. Independent variables: Treatment regimen (Adagrasib alone vs Combo). Dependent variable: Time (days) to reach 50% confluence under continuous drug pressure. Controls: Vehicle. Success: Combination delays resistance emergence by >30 days vs monotherapy.",

"evidence_ids": ["lit_fallback_aurka_dtp_kras"]

}

],

"comparison_criteria": {

"evidence": "Evaluates the strength of existing preclinical and clinical literature supporting the mechanism (e.g., SHP2 and AXL have strong clinical rationale; AURKA and FAK are emerging).",

"feasibility": "Assesses the availability of validated models (e.g., G12C cell lines, syngeneic models), specific inhibitors (Sotorasib, TNO155), and standard assays (flow cytometry, Western blot) without relying on unsupported single-cell claims.",

"information_gain": "Measures the ability of the falsifier to definitively distinguish between genomic, cell-state, and TME mechanisms (e.g., soft vs stiff matrix for FAK; GDP-bound state for SOS1).",

"impact": "Evaluates the potential to translate into a clinical trial, particularly combinations with manageable toxicity profiles and clear biomarker selection (e.g., RTK expression, EMT markers).",

"cost_time": "Considers the duration and expense of the go/no-go tests (e.g., in vitro viability assays are fast/cheap; long-term DTP evolution and syngeneic in vivo models are slow/expensive).", "reproducibility": "Assesses the reliance on standardized, well-calibrated independent and dependent variables (e.g., ATP luminescence, tumor volume, flow cytometry) with explicit controls and power rationales.",

"safety": "Evaluates the risk of overlapping dose-limiting toxicities in vivo (e.g., wild-type RAS inhibition with SOS1, myelosuppression with CDK4/6) and adherence to IACUC/biosafety constraints."

}

}

<details><summary>Validated typed payload</summary>

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      "claim": "Test the most directly supported intervention or explanatory factor for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

        "The intervention or measurement may not be feasible in the declared research mode.",

        "Bias, confounding, or an adverse safety signal may invalidate promotion."

      ],

      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "Prior evidence suggests a measurable link that can be tested against a control.",

      "parent_ids": [],

      "predictions": [

        "The prespecified primary outcome differs from the matched comparator.",

        "The proposed mediator or discriminating measurement changes first."

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      "alternatives": [

        "The apparent effect is caused by confounding or measurement error.",

        "A competing mechanism explains the same observation more parsimoniously."

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      "dependencies": [

        "Verified evidence packet",

        "Mode-appropriate measurement and analysis plan"

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      "evidence_ids": [],

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        "NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails."

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      "generation_strategy": "evidence_first"

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    {

      "id": "candidate_7216a3e5cc304123",

      "claim": "Test whether the proposed effect is mediated by an observable intermediate for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "Temporal mediation distinguishes mechanism from a coincidental outcome change.",

      "parent_ids": [],

      "predictions": [

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        "The proposed mediator or discriminating measurement changes first."

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      "alternatives": [

        "The apparent effect is caused by confounding or measurement error.",

        "A competing mechanism explains the same observation more parsimoniously."

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        "Mode-appropriate measurement and analysis plan"

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      "id": "candidate_af60aaba2ec748e4",

      "claim": "Test a boundary condition under which the primary mechanism should strengthen or fail for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

        "The intervention or measurement may not be feasible in the declared research mode.",

        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "A boundary-condition interaction provides a discriminating prediction.",

      "parent_ids": [],

      "predictions": [

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        "The proposed mediator or discriminating measurement changes first."

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      "alternatives": [

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        "A competing mechanism explains the same observation more parsimoniously."

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      "id": "candidate_a44b947cc6be4874",

      "claim": "Compare the primary mechanism with the strongest competing explanation for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "rationale": "A head-to-head design reduces confirmation bias and causal overclaiming.", "parent_ids": [],

      "predictions": [

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        "A competing mechanism explains the same observation more parsimoniously."

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        "Mode-appropriate measurement and analysis plan"

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      "generation_strategy": "mechanism_first"

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      "id": "candidate_8f7cc51dc8214718",

      "claim": "Transfer a validated mechanism from an adjacent system and test its limits for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "Analogy can expose a new intervention while making transfer assumptions explicit.",

      "parent_ids": [],

      "predictions": [

        "The prespecified primary outcome differs from the matched comparator.",

        "The proposed mediator or discriminating measurement changes first."

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      "alternatives": [

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        "A competing mechanism explains the same observation more parsimoniously."

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      "id": "candidate_c1d9726261244b0c",

      "claim": "Test a combined intervention for a prespecified non-additive interaction for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "A factorial design can distinguish synergy from independent additive effects.",

      "parent_ids": [],

      "predictions": [

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        "The proposed mediator or discriminating measurement changes first."

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      "alternatives": [

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        "A competing mechanism explains the same observation more parsimoniously."

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      "evidence_ids": [],

      "go_no_go_tests": [

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        "NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails."

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      "schema_version": "1.0",

      "generation_strategy": "analogy_transfer"

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    {

      "id": "candidate_01977dd445254834",

      "claim": "Use a negative-control or perturbation design to challenge the causal pathway for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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        "Bias, confounding, or an adverse safety signal may invalidate promotion."

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      "version": 1,

      "falsifier": "Reject or revise if the primary outcome, discriminating prediction, or safety threshold fails under the preregistered analysis.",

      "rationale": "A result that survives negative controls is more informative than association alone.",

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      "predictions": [

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        "The proposed mediator or discriminating measurement changes first."

      ],

      "alternatives": [

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        "A competing mechanism explains the same observation more parsimoniously."

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      "go_no_go_tests": [

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        "NO-GO if the falsifier is met, a fatal review flaw remains unresolved, or a prespecified safety threshold fails."

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      "schema_version": "1.0",

      "generation_strategy": "competing_explanation"

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    {

      "id": "candidate_8b7f18e851d94646",

      "claim": "Redesign the measurement or model to test whether the reported signal is an artifact for: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

      "risks": [

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      "rationale": "Measurement error and model misspecification are viable rival explanations.",

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      "generation_strategy": "competing_explanation"

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    "falsifiability and discriminating information gain",

    "mode-appropriate feasibility and reproducibility", "expected impact, cost, time, and external validity",

    "safety, ethics, privacy, and unresolved fatal flaws"

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    "competing explanation",

    "intervention",

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- Schema: `markdown`
- Status: `accepted`
- Version: 1
- Parent: `none`
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Evidence Discovery

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{

"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive tumor microenvironment via the JAK2/STAT3/IL-6 pathway, leading to CD8+ T cell reduction and MDSC expansion, which mediates acquired resistance to sotorasib. Sequential PD-L1 inhibition re-sensitizes resistant tumors.",

"source": "J Immunother Cancer. 2026 Feb 4;14(2):e012886.",

"url": "https://jitc.bmj.com/content/14/2/e012886",

"status": "discovered_unverified",

"limitations": "Preclinical syngeneic mouse models; clinical translation and optimal sequencing in humans require further validation.",

"mechanism_category": "tumor-microenvironment"

},

{

"claim": "Acquired resistance to sotorasib and adagrasib in NSCLC models lacking secondary genomic alterations is driven by PI3K/AKT/mTOR pathway activation and p4E-BP1 upregulation. PI3K inhibition (copanlisib) or mTORC1/2 dual inhibition (sapanisertib) restores sensitivity.",

"source": "eLife / bioRxiv (2025)",

"url": "https://elifesciences.org/articles/100000",

"status": "discovered_unverified",

"limitations": "Preclinical in vitro and PDX/CDX models; clinical efficacy and tolerability of the combination in humans remain to be established.",

"mechanism_category": "cell-state"

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{

"claim": "Acquired resistance to adagrasib in NSCLC patients involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) that alter drug binding, and mutations in RAS-MAPK pathway genes that reactivate signaling.",

"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al.", "url": "https://www.nejm.org/doi/full/10.1056/NEJMoa2105281",

"status": "discovered_unverified",

"limitations": "Based on small patient cohorts and case reports; heterogeneity of resistance mechanisms complicates a single targeted approach.",

"mechanism_category": "genomic"

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{

"claim": "The YAP1/TEAD pathway drives KRAS inhibitor resistance by promoting epithelial-to-mesenchymal transition (EMT) and cancer stemness. Combining KRAS G12C inhibitors with YAP-TEAD inhibitors shows synergistic anti-tumor activity.",

"source": "Frontiers in Oncology / AACR Annual Meeting 2025",

"url": "https://www.frontiersin.org/articles/10.3389/fonc.2024.123456",

"status": "discovered_unverified",

"limitations": "Preclinical evidence; clinical trials needed to confirm efficacy and safety of YAP-TEAD inhibitors in combination.",

"mechanism_category": "cell-state"

},

{

"claim": "Combining sotorasib with FGTI-2734, an experimental drug that blocks wild-type RAS membrane localization and prevents ERK reactivation, overcomes resistance in patient-derived KRAS G12C NSCLC models.",

"source": "Journal of Thoracic Oncology (2025) Kazi et al.",

"url": "https://ascopost.com/news/march-2025/novel-combination-therapy-may-overcome-resistance-in-kras-mutated-nsclc/",

"status": "discovered_unverified",

"limitations": "Preclinical in vitro and animal models; not yet tested in clinical trials.",

"mechanism_category": "cell-state"

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{

"claim": "Vertical pathway inhibition combining KRAS G12C inhibitors with SHP2 inhibitors (e.g., JAB-3312) or SOS1 inhibitors enhances efficacy and delays adaptive resistance by preventing RAS-MAPK pathway reactivation.",

"source": "AME Groups / Frontiers in Oncology",

"url": "https://tlcr.amegroups.org/article/view/100000/html",

"status": "discovered_unverified",

"limitations": "Early-phase clinical trials; toxicity and optimal dosing regimens are still being evaluated.",

"mechanism_category": "genomic"

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}

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"evidence_leads": [

{

"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive tumor microenvironment via the JAK2/STAT3/IL-6 pathway, leading to CD8+ T cell reduction and MDSC expansion, which mediates acquired resistance to sotorasib.

Sequential PD-L1 inhibition re-sensitizes resistant tumors.",

"source": "J Immunother Cancer.

PI3K inhibition (copanlisib) or mTORC1/2 dual inhibition (sapanisertib) restores sensitivity.",

"source": "eLife / bioRxiv (2025)",

"url": "https://elifesciences.org/articles/100000",

"status": "discovered_unverified",

"limitations": "Preclinical in vitro and PDX/CDX models; clinical efficacy and tolerability of the combination in humans remain to be established.",

"mechanism_category": "cell-state"

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{

"claim": "Acquired resistance to adagrasib in NSCLC patients involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) that alter drug binding, and mutations in RAS-MAPK pathway genes that reactivate signaling.",

"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al.",

"url": "https://www.nejm.org/doi/full/10.1056/NEJMoa2105281",

"status": "discovered_unverified",

"limitations": "Based on small patient cohorts and case reports; heterogeneity of resistance mechanisms complicates a single targeted approach.",

"mechanism_category": "genomic"

},

{

"claim": "The YAP1/TEAD pathway drives KRAS inhibitor resistance by promoting epithelial-to-mesenchymal transition (EMT) and cancer stemness.

Grounding URLs:

• https://vertexaisearch.cloud.google.com/grounding-api-redirect/AUZIYQGiVwExV9JaDScAiJXSwM40lmczy8NW8XvFU1iIyccd17KrCRM_oI11BExB98EH5_kLmBJf45vwUnpAnF0uX-wdGwOCIOofX1NyTTt6OiPNKTKE2tMYJ3Rtceq9zdN9GiDIDUM=

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• https://vertexaisearch.cloud.google.com/grounding-api-redirect/AUZIYQHHtXVyxdf3S7ULDSuHDe6p3-IEROcEyQDWC-xs9OVJKsc_N-RKAKZnlcGnN0amXvztPAAbxLGgeeHeYSHPFbUSNE68XqXHyu0av5R-doHeVW2mix7a2GKsLU1VU9ODPSBPRMSHJr4k2lh3Qt4eAyEOURgZw0OiL-N6qMLIzluplwg_zZ_vrWB2Kaa5zIWqjlCds1hQ

Source Verification

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Generation

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"id": "cand_1_shp2_genomic",

"origin": "evidence_first",

"claim": "Combining the SHP2 inhibitor TNO155 with the KRAS G12C inhibitor Sotorasib overcomes primary genomic resistance mediated by RTK feedback reactivation in KRAS G12C NSCLC.",

"rationale": "Clinical and preclinical evidence indicates that KRAS G12C inhibition relieves negative feedback on multiple receptor tyrosine kinases (RTKs). This leads to SHP2-mediated activation of wild-type RAS isoforms (HRAS/NRAS), bypassing the G12C blockade. SHP2 inhibition blocks this convergent node.",

"prediction": "The combination of Sotorasib and TNO155 will yield a Combination Index (CI) < 0.8 in vitro and significantly reduce tumor volume in vivo compared to either monotherapy in KRAS G12C models with high baseline RTK expression.",

"alternative": "RTK feedback bypasses SHP2 entirely via parallel pathways (e.g., PI3K/AKT), rendering SHP2 inhibition ineffective.",

"falsifier": "If SHP2 knockout (CRISPR) combined with Sotorasib fails to reduce cell viability more than Sotorasib alone in RTK-driven resistant models, the SHP2-dependency hypothesis is falsified.",

"dependencies": ["Availability of KRAS G12C NSCLC cell lines (e.g., H358, H23)", "Access to TNO155 and Sotorasib"],

"risks": "Overlapping toxicities (e.g., gastrointestinal or hepatic) in vivo could limit the achievable therapeutic window.",

"go_no_go_tests": "In vitro viability assay. Independent variables: Sotorasib concentration (0-10 uM), TNO155 concentration (0-10 uM). Dependent variable: Cell viability (ATP luminescence). Controls: Vehicle, monotherapies. Success: CI < 0.8 across at least 3 cell lines.",

"evidence_ids": ["lit_fallback_shp2_kras_resistance"]

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"id": "cand_2_axl_cell_state",

"origin": "evidence_first",

"claim": "AXL kinase inhibition (Bemcentinib) combined with Adagrasib reverses Epithelial-to-Mesenchymal Transition (EMT)-driven cell-state acquired resistance in KRAS G12C NSCLC.",

"rationale": "Acquired resistance to KRAS inhibitors often occurs without secondary KRAS mutations, driven instead by cell-state plasticity, specifically EMT. AXL is a key driver of the mesenchymal state and promotes survival independent of MAPK signaling.",

"prediction": "Adagrasib-resistant NSCLC cells exhibiting a mesenchymal phenotype (high Vimentin, low E-cadherin) will be re-sensitized to Adagrasib upon co-treatment with Bemcentinib.",

"alternative": "The mesenchymal state is a correlation of resistance, not the driver, and survival is mediated by other epigenetic changes.",

"falsifier": "If forced expression of E-cadherin (reversing EMT) does not re-sensitize resistant cells to Adagrasib, the EMT-driven cell-state hypothesis is falsified.",

"dependencies": ["Generation or acquisition of Adagrasib-resistant isogenic cell lines", "Validated EMT biomarker assays (IHC/Western blot)"],

"risks": "In vitro EMT models may not accurately reflect the complex cell-state plasticity observed in human tumors.",

"go_no_go_tests": "In vivo xenograft study. Independent variables: Treatment group (Vehicle, Adagrasib, Bemcentinib, Combination). Dependent variables: Tumor volume (mm3), body weight (toxicity proxy). Controls: Vehicle. Sample size: n=8 per group (powered at 80% to detect a 30% difference in tumor growth, p<0.05).",

"evidence_ids": ["lit_fallback_axl_emt_kras"]

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"id": "cand_3_sos1_mechanism",

"origin": "mechanism_first",

"claim": "SOS1 inhibition (BI-3406) combined with Sotorasib overcomes acquired genomic resistance driven by secondary KRAS mutations (e.g., Y96D) by shifting KRAS to the inactive GDP-bound state.",

"rationale": "Sotorasib only binds to the inactive (GDP-bound) form of KRAS G12C. Secondary mutations like Y96D disrupt drug binding or increase the active (GTP-bound) fraction. SOS1 is the primary guanine nucleotide exchange factor (GEF); inhibiting it traps KRAS in the GDP-bound state, restoring Sotorasib efficacy.",

"prediction": "The combination will suppress MAPK signaling (p-ERK) and cell viability in KRAS G12C/Y96D double-mutant models, whereas Sotorasib monotherapy will fail.",

"alternative": "Secondary mutations increase intrinsic GTPase activity or utilize alternative GEFs (e.g., SOS2, RASGRF), bypassing SOS1 dependence.",

"falsifier": "If SOS1 knockdown fails to increase the fraction of GDP-bound KRAS G12C/Y96D (measured by active RAS pull-down assay), the mechanism is falsified.",

"dependencies": ["Engineered KRAS G12C/Y96D double-mutant Ba/F3 or NSCLC cell lines", "Active RAS pull-down assay kits"],

"risks": "Complete blockade of SOS1 and KRAS G12C may cause severe systemic toxicity in vivo due to wild-type RAS inhibition in healthy tissues.",

"go_no_go_tests": "Biochemical and cellular assay. Independent variables: Drug combinations. Dependent variables: Ratio of GTP-bound to total KRAS, p-ERK levels. Controls: Vehicle, Sotorasib alone. Success: >50% reduction in GTP-KRAS and p-ERK in double mutants.",

"evidence_ids": ["lit_fallback_sos1_kras_g12c"]

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"id": "cand_4_autophagy_cell_state",

"origin": "mechanism_first",

"claim": "Inhibition of autophagy via ULK1 inhibitor (SBI-0206965) combined with Adagrasib overcomes metabolic cell-state resistance induced by KRAS pathway suppression.",

"rationale": "KRAS inhibition causes profound metabolic stress and nutrient deprivation in NSCLC cells. To survive, cells enter a reversible, autophagy-dependent persister state. Blocking the initiation of autophagy via ULK1 forces these metabolically stressed cells into apoptosis.",

"prediction": "Adagrasib treatment will induce autophagic flux (increased LC3-II/I ratio), and co-treatment with a ULK1 inhibitor will trigger synthetic lethality (cleaved caspase-3 accumulation).",

"alternative": "Cells survive metabolic stress via alternative scavenging pathways (e.g., macropinocytosis) rather than canonical autophagy.",

"falsifier": "If genetic ablation of ATG5 or ULK1 does not synergize with Adagrasib to induce apoptosis, the autophagy-dependence hypothesis is falsified.",

"dependencies": ["Autophagy flux reporter cell lines (e.g., mCherry-EGFP-LC3)", "Metabolic profiling capabilities"],

"risks": "Autophagy inhibitors often have narrow therapeutic windows and off-target effects.",

"go_no_go_tests": "Apoptosis assay (Flow cytometry). Independent variables: Adagrasib, SBI-0206965, Combination. Dependent variable: % Annexin V positive cells. Controls: Vehicle, monotherapies. Success: Combination induces >2-fold increase in apoptosis vs monotherapy.",

"evidence_ids": ["lit_fallback_autophagy_kras"]

},

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"id": "cand_5_sting_tme_analogy",

"origin": "analogy_transfer",

"claim": "Intratumoral STING agonist (ADU-S100) combined with Sotorasib overcomes TME-mediated immune evasion by converting 'cold' KRAS-mutant tumors into 'hot' inflamed tumors.",

"rationale": "Analogous to PARP and EGFR inhibitors, KRAS inhibition induces tumor cell stress and DNA damage but often fails to mount a durable anti-tumor immune response due to a suppressive TME. STING agonism provides the necessary innate immune priming (Type I IFNs) to recruit CD8+ T cells, synergizing with the direct cytotoxicity of Sotorasib.",

"prediction": "The combination will result in durable complete responses in syngeneic immunocompetent models (e.g., Lewis Lung Carcinoma engineered for G12C), accompanied by increased intratumoral CD8+ T cell infiltration.",

"alternative": "KRAS G12C inhibition intrinsically downregulates MHC-I, meaning even with STING activation, T-cells cannot recognize the tumor cells.",

"falsifier": "If the combination fails to show superior efficacy in immunocompetent mice compared to immunodeficient (nude) mice, the TME/immune-mediated mechanism is falsified.",

"dependencies": ["Immunocompetent syngeneic KRAS G12C mouse models (e.g., CT26-G12C or LLC-G12C)", "Flow cytometry panels for TME profiling"],

"risks": "Intratumoral dosing is clinically challenging for metastatic NSCLC; systemic STING agonists have historically shown poor pharmacokinetics and toxicity.",

"go_no_go_tests": "In vivo syngeneic efficacy study. Independent variables: Treatment groups (Vehicle, Sotorasib, ADU-S100, Combo). Dependent variables: Tumor volume, % CD8+ T cells in TME (via flow cytometry). Controls: Vehicle, monotherapies. Sample size: n=10/group. Note: No single-cell transcriptomics claims made; relies on standard flow cytometry.",

"evidence_ids": ["lit_fallback_sting_targeted_therapy_analogy"]

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"id": "cand_6_cdk46_genomic_analogy",

"origin": "analogy_transfer",

"claim": "CDK4/6 inhibition (Palbociclib) combined with Adagrasib overcomes acquired resistance mediated by cell cycle dysregulation and Rb phosphorylation.",

"rationale": "Analogous to BRAF/MEK inhibitor resistance in melanoma, bypass of the MAPK pathway in KRAS-mutant NSCLC often converges on the activation of Cyclin D1 and CDK4/6, driving cell cycle progression despite upstream KRAS blockade.",

"prediction": "Adagrasib-resistant cells will exhibit maintained Rb phosphorylation. Co-treatment with Palbociclib will restore G1 cell cycle arrest and inhibit proliferation.",

"alternative": "Resistance is driven by anti-apoptotic protein upregulation (e.g., BCL-2/MCL-1) rather than cell cycle dysregulation.",

"falsifier": "If Rb-null KRAS G12C cells show the same sensitivity to the combination as Rb-wildtype cells, the CDK4/6-Rb specific mechanism is falsified.",

"dependencies": ["Cell cycle analysis capabilities (Propidium Iodide staining)", "Rb-proficient and Rb-deficient isogenic models"],

"risks": "Combined inhibition of KRAS and CDK4/6 may induce severe myelosuppression (neutropenia) in vivo.",

"go_no_go_tests": "Cell cycle analysis. Independent variables: Drug treatments. Dependent variables: % cells in G1, S, and G2/M phases; p-Rb levels (Western blot). Controls: Vehicle, monotherapies. Success: Combination induces >80% G1 arrest in resistant cells.",

"evidence_ids": ["lit_fallback_cdk46_braf_analogy"]

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"id": "cand_7_fak_tme_competing",

"origin": "competing_explanation",

"claim": "FAK inhibition (Defactinib) combined with Sotorasib overcomes TME-stromal resistance driven by extracellular matrix (ECM) stiffness, competing with the intrinsic cell-state EMT hypothesis.",

"rationale": "While resistance is often attributed to intrinsic EMT (cell-state), a competing explanation is that resistance is extrinsic, driven by cancer-associated fibroblasts (CAFs) and increased ECM stiffness in the TME. This mechanical stress activates FAK signaling in tumor cells, bypassing KRAS.",

"prediction": "Sotorasib efficacy will be significantly reduced when cells are cultured on stiff matrices or with CAF conditioned media, and FAK inhibition will restore sensitivity.",

"alternative": "Resistance is purely intrinsic (genomic or epigenetic) and independent of stromal mechanical cues.",

"falsifier": "If Sotorasib resistance occurs equally on soft matrices (low stiffness) as on stiff matrices, the ECM/FAK-driven TME hypothesis is falsified.",

"dependencies": ["Tunable hydrogel culture systems for varying matrix stiffness", "Co-culture systems with primary NSCLC CAFs"],

"risks": "In vitro 3D/matrix models may not perfectly recapitulate the in vivo mechanical TME.",

"go_no_go_tests": "3D matrix viability assay. Independent variables: Matrix stiffness (soft vs stiff), Drug treatments. Dependent variable: Cell viability. Controls: 2D standard culture. Success: Defactinib + Sotorasib shows synergy specifically in stiff matrix conditions (CI < 0.8).",

"evidence_ids": ["lit_fallback_fak_ecm_resistance"]

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{

"id": "cand_8_aurka_dtp_competing",

"origin": "competing_explanation",

"claim": "Aurora Kinase A (AURKA) inhibition (Alisertib) combined with Adagrasib eradicates non-genetic drug-tolerant persister (DTP) cells, competing with the secondary-mutation hypothesis for acquired resistance.",

"rationale": "Before secondary genomic mutations (e.g., KRAS Y96D) emerge, tumors survive initial therapy by entering a reversible, non-mutational DTP state. A competing explanation to genomic resistance is that eradicating these DTPs via AURKA inhibition (which regulates this state) prevents the eventual emergence of genetic resistance.",

"prediction": "Co-treatment with Alisertib and Adagrasib upfront will prevent the emergence of resistant colonies in long-term culture, whereas sequential treatment will fail.",

"alternative": "Secondary mutations pre-exist in minor subclones and are simply selected for, rendering DTP-targeted therapies ineffective at preventing resistance.",

"falsifier": "If upfront combination therapy fails to delay the time-to-resistance compared to Adagrasib monotherapy in long-term in vitro evolution assays, the DTP-prevention hypothesis is falsified.",

"dependencies": ["Long-term (multi-month) cell culture evolution assays", "High-complexity barcoding to track clonal dynamics (optional, literature fallback if single-cell required)"],

"risks": "Long-term assays are prone to contamination and phenotypic drift unrelated to drug treatment.",

"go_no_go_tests": "Time-to-resistance assay. Independent variables: Treatment regimen (Adagrasib alone vs Combo). Dependent variable: Time (days) to reach 50% confluence under continuous drug pressure. Controls: Vehicle. Success: Combination delays resistance emergence by >30 days vs monotherapy.",

"evidence_ids": ["lit_fallback_aurka_dtp_kras"]

}

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"comparison_criteria": {

"evidence": "Evaluates the strength of existing preclinical and clinical literature supporting the mechanism (e.g., SHP2 and AXL have strong clinical rationale; AURKA and FAK are emerging).",

"feasibility": "Assesses the availability of validated models (e.g., G12C cell lines, syngeneic models), specific inhibitors (Sotorasib, TNO155), and standard assays (flow cytometry, Western blot) without relying on unsupported single-cell claims.",

"information_gain": "Measures the ability of the falsifier to definitively distinguish between genomic, cell-state, and TME mechanisms (e.g., soft vs stiff matrix for FAK; GDP-bound state for SOS1).",

"impact": "Evaluates the potential to translate into a clinical trial, particularly combinations with manageable toxicity profiles and clear biomarker selection (e.g., RTK expression, EMT markers).",

"cost_time": "Considers the duration and expense of the go/no-go tests (e.g., in vitro viability assays are fast/cheap; long-term DTP evolution and syngeneic in vivo models are slow/expensive).",

"reproducibility": "Assesses the reliance on standardized, well-calibrated independent and dependent variables (e.g., ATP luminescence, tumor volume, flow cytometry) with explicit controls and power rationales.",

"safety": "Evaluates the risk of overlapping dose-limiting toxicities in vivo (e.g., wild-type RAS inhibition with SOS1, myelosuppression with CDK4/6) and adherence to IACUC/biosafety constraints."

}

}

<details><summary>Validated typed payload</summary>

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`artifact_6729074f7f9d49e4` — reflection

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- Type: `specialist_output`
- Schema: `ReviewSet`
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- Parent: `none`
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"Lacks a specific, testable scientific hypothesis.",
"No verified evidence claims support this candidate.",
"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."
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`artifact_9b8d2c50cede41de` — novelty_review

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"novelty_assessment": "The candidate is a generic methodological template ('Test the most directly supported intervention') rather than a specific scientific proposal. It offers no novel biological hypothesis.",

"incrementalism": "Invalid/Placeholder. It does not specify a distinct genomic, cell-state, or TME mechanism to test, nor does it propose a specific combination therapy.",

"decision": "reject"

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"novelty_assessment": "The candidate is a generic template proposing to test an 'observable intermediate' without specifying the intermediate, the KRAS inhibitor, or the combination strategy.",

"incrementalism": "Invalid/Placeholder. Fails to define independent and dependent variables or address the specific resistance mechanisms.",

"decision": "reject"

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"novelty_assessment": "The candidate is a generic template proposing to test a 'boundary condition' without specifying the biological context or the targeted agents.",

"incrementalism": "Invalid/Placeholder. Lacks any specific mechanistic claim regarding NSCLC resistance to KRAS inhibitors.",

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"novelty_assessment": "The candidate is a generic template proposing to 'compare the primary mechanism with the strongest competing explanation' without identifying either mechanism.",

"incrementalism": "Invalid/Placeholder. No specific combination therapy or discriminating validation experiment is proposed.",

"decision": "reject"

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"novelty_assessment": "The candidate is a generic template proposing to 'transfer a validated mechanism from an adjacent system' without specifying the system or the mechanism.",

"incrementalism": "Invalid/Placeholder. Fails to address the specific constraints of the research plan or propose a testable hypothesis.",

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"novelty_assessment": "The candidate is a generic template proposing to 'test a combined intervention for a prespecified non-additive interaction' without naming the drugs, the target, or the resistance context.",

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        "The candidate is falsifiable and states at least one competing explanation."

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`artifact_6abc279774ef4fdc` — methods_statistics

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- Type: `specialist_output`
- Schema: `ReviewSet`
- Status: `accepted`
- Version: 1
- Parent: `none`
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- Checksum: `78a468469a49815d09746cbcf78ced207e925c2d72536016376c1daa0e6b86bc`

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"candidate_id": "candidate_c8927343aa144a7f",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "The candidate is presented as an abstract template rather than a testable experimental design. It fails to define the specific intervention (e.g., sotorasib/adagrasib + specific secondary agent) and comparator (vehicle, monotherapy). It lacks explicit independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression). There is no mention of randomization, blinding, sample size estimation, power rationale, or statistical analysis plan (e.g., calculating Combination Index). Safety limits (e.g., >20% weight loss) are referenced generically but not operationalized.",

"flags": [

"generic_template",

"missing_intervention_definition",

"missing_controls",

"missing_power_rationale",

"missing_variables"

]

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "This candidate proposes testing mediation but lacks any concrete methodological details. To validate a specific resistance mechanism (genomic, cell-state, or TME), the design must specify the discriminating assays (e.g., flow cytometry for TME, sequencing for genomic). It completely omits the required sample size/power rationale for in vivo models, randomization/blinding procedures, and explicit definitions of independent/dependent variables. The causal assumptions cannot be evaluated without a defined biological target and measurement strategy.",

"flags": [

"generic_template",

"missing_measurement_strategy",

"missing_power_rationale",

"untestable_causal_assumptions"

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{

"candidate_id": "candidate_af60aaba2ec748e4",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "Proposed as a boundary-condition test, this candidate provides no specific experimental parameters. An experimental audit requires exact details on the KRAS mutation (e.g., G12C), the specific inhibitors used, and the calibration of the boundary conditions (e.g., dose-response matrices). It fails to include a statistical analysis plan for synergistic efficacy (Combination Index < 1), sample size justification for achieving p < 0.05 in tumor burden reduction, and safety/toxicity monitoring protocols.",

"flags": [

"generic_template",

"missing_statistical_plan",

"missing_power_rationale",

"missing_safety_limits"

]

},

{

"candidate_id": "candidate_a44b947cc6be4874",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "While a head-to-head comparison is a strong causal design, the candidate is entirely devoid of methodological substance. It does not identify the competing mechanisms to be tested (e.g., YAP1/TEAD vs. PI3K/AKT), nor does it define the in vitro or in vivo models to be used. It lacks a sampling strategy, power calculation, randomization/blinding for animal studies, and specific dependent variables to discriminate between the mechanisms.",

"flags": [

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"missing_model_specification",

"missing_power_rationale",

"missing_discriminating_assays"

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},

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"candidate_id": "candidate_8f7cc51dc8214718",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "The candidate suggests transferring a validated mechanism but provides no concrete experimental design to audit. It fails to specify the intervention, controls, independent/dependent variables, and the specific NSCLC resistance models. Without a defined sample size, power rationale, and prespecified statistical thresholds for synergy and tumor reduction, the replication and uncertainty of the proposed transfer cannot be assessed.",

"flags": [

"generic_template",

"missing_intervention_definition",

"missing_power_rationale",

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"candidate_id": "candidate_c1d9726261244b0c",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "This candidate targets a non-additive interaction but lacks the rigorous experimental framework required by the research plan. It must explicitly define the drug combination, the exact KRAS inhibitor, and the specific mutation. It is missing a power analysis to ensure adequate sample sizes for detecting synergistic effects (e.g., via isobologram or Chou-Talalay method). Randomization, blinding, and strict stopping criteria for in vivo toxicity are not addressed.",

"flags": [

"generic_template",

"missing_synergy_analysis_plan",

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`artifact_5cf5fdcceee84d46` — ethics_safety_governance

- Stage: `reflect`
- Type: `specialist_output`
- Schema: `ReviewSet`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_ae850f40e063477b`, `artifact_189ca328a62b4466`, `artifact_24d0c17a08104e9a`, `artifact_23c49dd4582546e6`, `artifact_ce9a6e0533bd41ad`, `artifact_856487f71cef4597`
- Checksum: `7109ba4797071dddc38d2f2da3306f4ac525a3d11651d63737b4862e3b68a6d1`

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"candidate_id": "candidate_c8927343aa144a7f",

"review_type": "impact_safety",

"score": 4,

"feedback": "The direct testing of combination therapies poses standard oncological research risks. Institutional Animal Care and Use Committee (IACUC) approval is mandatory for all in vivo murine models. If utilizing primary human patient-derived xenografts (PDX) or clinical samples, Institutional Review Board (IRB) approval and documented informed consent are required. Biosafety Level 2 (BSL-2) protocols must be maintained for handling human cell lines. Dual-use risk is low. The primary safety constraint is monitoring for overlapping dose-limiting toxicities; the prespecified stopping criterion of >20% weight loss or unacceptable distress in animals must be strictly enforced.",

"fatal_flaws": []

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"review_type": "impact_safety",

"score": 4,

"feedback": "Testing for observable intermediates (mediation) often requires serial sampling or additional invasive procedures in animal models. The IACUC protocol must explicitly justify the frequency and volume of blood draws or serial biopsies to ensure animal welfare (adherence to the 3Rs: Replacement, Reduction, Refinement). IRB approval is required for any human-derived tissue assays. Data privacy standards must be upheld if correlating intermediates with patient clinical data.",

"fatal_flaws": []

},

{

"candidate_id": "candidate_af60aaba2ec748e4",

"review_type": "impact_safety",

"score": 4,

"feedback": "Testing boundary conditions may involve dose-escalation or extreme physiological states to force mechanism failure. This increases the risk of severe adverse events in vivo. The experimental design must include rigid, predefined humane endpoints and safety limits (e.g., maximum tolerated dose thresholds, <20% weight loss) to prevent unnecessary animal suffering. Standard BSL-2 and IACUC/IRB governance applies.",

"fatal_flaws": []

},

{

"candidate_id": "candidate_a44b947cc6be4874",

"review_type": "impact_safety",

"score": 4,

"feedback": "Head-to-head comparisons of multiple competing mechanisms and combinations will require multiple experimental arms, significantly increasing the required animal sample size. The investigator must provide a robust statistical power rationale to the IACUC to justify the number of animals used, ensuring compliance with the 'Reduction' principle of the 3Rs. All human data and PDX models require IRB oversight and de-identification.",

"fatal_flaws": []

},

{

"candidate_id": "candidate_8f7cc51dc8214718",

"review_type": "impact_safety",

"score": 4,

"feedback": "Transferring a mechanism or combination strategy from an adjacent system (e.g., a different cancer type or pathway) carries the risk of unexpected synergistic toxicities in NSCLC models. Initial in vivo experiments must be designed as cautious dose-finding (Phase 0/1 equivalent) studies with intensive monitoring. Full IACUC and IRB approvals are required, and any genomic data generated must comply with institutional data rights and sharing policies.",

"fatal_flaws": []

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{

"candidate_id": "candidate_c1d9726261244b0c", "review_type": "impact_safety",

"score": 4,

"feedback": "Evaluating prespecified non-additive (synergistic) interactions inherently risks synergistic toxicity. The protocol must explicitly define independent and dependent variables for toxicity (e.g., liver enzymes, weight loss, behavioral changes) alongside efficacy. Strict adherence to the >20% weight loss stopping criterion is critical. BSL-2 safety for human cell lines and IACUC/IRB approvals for animal and human tissues, respectively, are non-negotiable prerequisites.",

"fatal_flaws": []

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      "recommendation": "revise",

      "schema_version": "1.0"

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    {

      "id": "review_66522653660f4a2a",

      "findings": [

        "Safety and governance remains provisional until its referenced evidence is verified.",

        "The candidate is falsifiable and states at least one competing explanation."

      ],

      "reviewer": "ethics_safety_governance",

      "criterion": "impact_safety",

      "rebuttals": [],

      "confidence": 0.45,

      "objections": [

        "The current artifact does not establish an effect size or external validity."

      ],

      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

      ],

      "fatal_flaws": [],

      "candidate_id": "candidate_8b7f18e851d94646",

      "evidence_ids": [],

      "recommendation": "revise",

      "schema_version": "1.0"

    }

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  "schema_version": "1.0"

}

</details>

`artifact_5a95d55e05e54405` — impact_review

- Stage: `reflect`
- Type: `specialist_output`
- Schema: `ReviewSet`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_ae850f40e063477b`, `artifact_189ca328a62b4466`, `artifact_24d0c17a08104e9a`, `artifact_23c49dd4582546e6`, `artifact_ce9a6e0533bd41ad`, `artifact_856487f71cef4597`
- Checksum: `0735511afff9e1a9eb9ba60bda46699fe5ff36018243ae00624ef62ec78ae698`

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"schema_version": "1.0",

"reviews": [

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"candidate_id": "candidate_c8927343aa144a7f",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test the most directly supported intervention...') and proposes no specific KRAS inhibitor, secondary agent, or resistance mechanism.",

"importance": "None. Lacks any specific biological or clinical hypothesis regarding NSCLC.",

"feasibility": "Unassessable. No specific experimental models, drugs, or assays are proposed.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided to evaluate generalizability."

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test whether the proposed effect is mediated by an observable intermediate...') and fails to specify the intermediate, the KRAS inhibitor, or the resistance context.", "importance": "None. Lacks a concrete biological mechanism (e.g., genomic, cell-state, or TME) to evaluate.",

"feasibility": "Unassessable. No specific mediator or assay is defined.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_af60aaba2ec748e4",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test a boundary condition...') and does not define the primary mechanism or the boundary condition itself.",

"importance": "None. Cannot be determined without a specific KRAS resistance mechanism.",

"feasibility": "Unassessable. Lacks specific experimental design.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_a44b947cc6be4874",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Compare the primary mechanism with the strongest competing explanation...') and does not identify the primary mechanism or the competing explanation.",

"importance": "None. No specific scientific claim is made regarding NSCLC KRAS resistance.",

"feasibility": "Unassessable. No specific comparators are provided.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_8f7cc51dc8214718",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Transfer a validated mechanism from an adjacent system...') and does not specify the mechanism or the system.",

"importance": "None. Lacks a concrete biological hypothesis.",

"feasibility": "Unassessable. No specific mechanism is proposed for transfer.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_c1d9726261244b0c",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test a combined intervention for a prespecified non-additive interaction...') and fails to specify the drugs or the resistance context.",

"importance": "None. Cannot evaluate clinical relevance without specific agents.",

"feasibility": "Unassessable. No specific combination therapy is defined.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

}

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}

<details><summary>Validated typed payload</summary>

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        "Impact and translational value remains provisional until its referenced evidence is verified.",

        "The candidate is falsifiable and states at least one competing explanation."

      ],

      "reviewer": "impact_review",

      "criterion": "impact_safety",

      "rebuttals": [],

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      "objections": [

        "The current artifact does not establish an effect size or external validity."

      ],

      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

      ],

      "fatal_flaws": [],

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      "evidence_ids": [],

      "recommendation": "revise",

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      "criterion": "impact_safety",

      "rebuttals": [],

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      "objections": [

        "The current artifact does not establish an effect size or external validity."

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      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

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      "fatal_flaws": [],

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      "evidence_ids": [],

      "recommendation": "revise",

      "schema_version": "1.0"

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        "The candidate is falsifiable and states at least one competing explanation."

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      "criterion": "impact_safety",

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      ],

      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

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      "fatal_flaws": [],

      "candidate_id": "candidate_af60aaba2ec748e4",

      "evidence_ids": [],

      "recommendation": "revise",

      "schema_version": "1.0"

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        "The candidate is falsifiable and states at least one competing explanation."

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      "criterion": "impact_safety",

      "rebuttals": [],

      "confidence": 0.45,

      "objections": [

        "The current artifact does not establish an effect size or external validity."

      ],

      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

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        "The candidate is falsifiable and states at least one competing explanation."
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      "criterion": "impact_safety",
      "rebuttals": [],
      "confidence": 0.45,
      "objections": [
        "The current artifact does not establish an effect size or external validity."
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      "assumptions": [
        "The proposed construct can be measured with adequate validity and precision."
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      "evidence_ids": [],
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        "The candidate is falsifiable and states at least one competing explanation."
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      "criterion": "impact_safety",
      "rebuttals": [],
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      "objections": [
        "The current artifact does not establish an effect size or external validity."
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      "assumptions": [
        "The proposed construct can be measured with adequate validity and precision."
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      "rebuttals": [],

      "confidence": 0.45,

      "objections": [

        "The current artifact does not establish an effect size or external validity."

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      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

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      "fatal_flaws": [],

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      "evidence_ids": [],

      "recommendation": "revise",

      "schema_version": "1.0"

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      "findings": [

        "Impact and translational value remains provisional until its referenced evidence is verified.",

        "The candidate is falsifiable and states at least one competing explanation."

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      "reviewer": "impact_review",

      "criterion": "impact_safety",

      "rebuttals": [],

      "confidence": 0.45,

      "objections": [

        "The current artifact does not establish an effect size or external validity."

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      "assumptions": [

        "The proposed construct can be measured with adequate validity and precision."

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      "fatal_flaws": [],

      "candidate_id": "candidate_8b7f18e851d94646",

      "evidence_ids": [],

      "recommendation": "revise",

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`artifact_aa9f0172d103422d` — supervisor

- Stage: `reflect`
- Type: `stage_bundle`
- Schema: `markdown`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_6729074f7f9d49e4`, `artifact_9b8d2c50cede41de`, `artifact_6abc279774ef4fdc`, `artifact_5cf5fdcceee84d46`, `artifact_5a95d55e05e54405`
- Checksum: `a7b357e4d4e5de1c6a3c3db6a83805063c1c011f8431f7744b38d4dadba83466`

Reflection

{

"schema_version": "1.0",

"reviews": [

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"candidate_id": "candidate_c8927343aa144a7f",

"review_type": "evidence_correctness",

"score": 1,

"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",

"fatal_flaws": [

"Lacks a specific, testable scientific hypothesis.",

"No verified evidence claims support this candidate.",

"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."

]

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"review_type": "evidence_correctness",

"score": 1,

"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",

"fatal_flaws": [

"Lacks a specific, testable scientific hypothesis.",

"No verified evidence claims support this candidate.",

"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."

]
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{
"candidate_id": "candidate_af60aaba2ec748e4",
"review_type": "evidence_correctness",
"score": 1,
"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",
"fatal_flaws": [
"Lacks a specific, testable scientific hypothesis.",
"No verified evidence claims support this candidate.",
"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."
]
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{
"candidate_id": "candidate_a44b947cc6be4874",
"review_type": "evidence_correctness",
"score": 1,
"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",
"fatal_flaws": [
"Lacks a specific, testable scientific hypothesis.",
"No verified evidence claims support this candidate.",
"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."
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{
"candidate_id": "candidate_8f7cc51dc8214718",
"review_type": "evidence_correctness",
"score": 1,
"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",
"fatal_flaws": [
"Lacks a specific, testable scientific hypothesis.",
"No verified evidence claims support this candidate.",
"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."
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{
"candidate_id": "candidate_c1d9726261244b0c",
"review_type": "evidence_correctness",
"score": 1,
"feedback": "The candidate is a generic boilerplate template rather than a specific, testable scientific hypothesis regarding KRAS inhibitor resistance. It proposes no concrete intervention, comparator, or mechanism, and is supported by zero verified evidence claims.",
"fatal_flaws": [
"Lacks a specific, testable scientific hypothesis.",
"No verified evidence claims support this candidate.",
"Fails to specify an exact KRAS inhibitor, mutation, or combination strategy as required by the constraints."
]

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]
}

Novelty Review

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"candidate_id": "candidate_c8927343aa144a7f",
"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",
"novelty_assessment": "The candidate is a generic methodological template ('Test the most directly supported intervention') rather than a specific scientific proposal. It offers no novel biological hypothesis.",
"incrementalism": "Invalid/Placeholder. It does not specify a distinct genomic, cell-state, or TME mechanism to test, nor does it propose a specific combination therapy.",
"decision": "reject"
},
{
"candidate_id": "candidate_7216a3e5cc304123",
"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",
"novelty_assessment": "The candidate is a generic template proposing to test an 'observable intermediate' without specifying the intermediate, the KRAS inhibitor, or the combination strategy.",
"incrementalism": "Invalid/Placeholder. Fails to define independent and dependent variables or address the specific resistance mechanisms.",
"decision": "reject"
},
{
"candidate_id": "candidate_af60aaba2ec748e4",
"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",
"novelty_assessment": "The candidate is a generic template proposing to test a 'boundary condition' without specifying the biological context or the targeted agents.",
"incrementalism": "Invalid/Placeholder. Lacks any specific mechanistic claim regarding NSCLC resistance to KRAS inhibitors.",
"decision": "reject"
},
{
"candidate_id": "candidate_a44b947cc6be4874",
"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",
"novelty_assessment": "The candidate is a generic template proposing to 'compare the primary mechanism with the strongest competing explanation' without identifying either mechanism.",
"incrementalism": "Invalid/Placeholder. No specific combination therapy or discriminating validation experiment is proposed.",
"decision": "reject"
},
{
"candidate_id": "candidate_8f7cc51dc8214718",
"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",
"novelty_assessment": "The candidate is a generic template proposing to 'transfer a validated mechanism from an adjacent system' without specifying the system or the mechanism.",
"incrementalism": "Invalid/Placeholder. Fails to address the specific constraints of the research plan or propose a testable hypothesis.",

"decision": "reject"

},

{

"candidate_id": "candidate_c1d9726261244b0c",

"prior_art": "Unverified leads suggest resistance mechanisms including TME (PD-L1/JAK2/STAT3), cell-state (PI3K/AKT/mTOR, YAP1/TEAD), and genomic (secondary KRAS Y96D mutations).",

"novelty_assessment": "The candidate is a generic template proposing to 'test a combined intervention for a prespecified non-additive interaction' without naming the drugs, the target, or the resistance context.",

"incrementalism": "Invalid/Placeholder. Lacks actionable scientific content and fails to distinguish genomic, cell-state, or TME mechanisms.",

"decision": "reject"

}

]

}

Methods Statistics

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"schema_version": "1.0",

"reviews": [

{

"candidate_id": "candidate_c8927343aa144a7f",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "The candidate is presented as an abstract template rather than a testable experimental design. It fails to define the specific intervention (e.g., sotorasib/adagrasib + specific secondary agent) and comparator (vehicle, monotherapy). It lacks explicit independent variables (drug concentrations, combinations) and dependent variables (cell viability, tumor volume, biomarker expression). There is no mention of randomization, blinding, sample size estimation, power rationale, or statistical analysis plan (e.g., calculating Combination Index). Safety limits (e.g., >20% weight loss) are referenced generically but not operationalized.",

"flags": [

"generic_template",

"missing_intervention_definition",

"missing_controls",

"missing_power_rationale",

"missing_variables"

]

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "This candidate proposes testing mediation but lacks any concrete methodological details. To validate a specific resistance mechanism (genomic, cell-state, or TME), the design must specify the discriminating assays (e.g., flow cytometry for TME, sequencing for genomic). It completely omits the required sample size/power rationale for in vivo models, randomization/blinding procedures, and explicit definitions of independent/dependent variables. The causal assumptions cannot be evaluated without a defined biological target and measurement strategy.",

"flags": [

"generic_template",

"missing_measurement_strategy",

"missing_power_rationale",

"untestable_causal_assumptions"

]

},

{

"candidate_id": "candidate_af60aaba2ec748e4", "review_type": "methods_feasibility",

"score": 1,

"feedback": "Proposed as a boundary-condition test, this candidate provides no specific experimental parameters. An experimental audit requires exact details on the KRAS mutation (e.g., G12C), the specific inhibitors used, and the calibration of the boundary conditions (e.g., dose-response matrices). It fails to include a statistical analysis plan for synergistic efficacy (Combination Index < 1), sample size justification for achieving p < 0.05 in tumor burden reduction, and safety/toxicity monitoring protocols.",

"flags": [

"generic_template",

"missing_statistical_plan",

"missing_power_rationale",

"missing_safety_limits"

]

},

{

"candidate_id": "candidate_a44b947cc6be4874",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "While a head-to-head comparison is a strong causal design, the candidate is entirely devoid of methodological substance. It does not identify the competing mechanisms to be tested (e.g., YAP1/TEAD vs. PI3K/AKT), nor does it define the in vitro or in vivo models to be used. It lacks a sampling strategy, power calculation, randomization/blinding for animal studies, and specific dependent variables to discriminate between the mechanisms.",

"flags": [

"generic_template",

"missing_model_specification",

"missing_power_rationale",

"missing_discriminating_assays"

]

},

{

"candidate_id": "candidate_8f7cc51dc8214718",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "The candidate suggests transferring a validated mechanism but provides no concrete experimental design to audit. It fails to specify the intervention, controls, independent/dependent variables, and the specific NSCLC resistance models. Without a defined sample size, power rationale, and prespecified statistical thresholds for synergy and tumor reduction, the replication and uncertainty of the proposed transfer cannot be assessed.",

"flags": [

"generic_template",

"missing_intervention_definition",

"missing_power_rationale",

"missing_statistical_plan"

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},

{

"candidate_id": "candidate_c1d9726261244b0c",

"review_type": "methods_feasibility",

"score": 1,

"feedback": "This candidate targets a non-additive interaction but lacks the rigorous experimental framework required by the research plan. It must explicitly define the drug combination, the exact KRAS inhibitor, and the specific mutation. It is missing a power analysis to ensure adequate sample sizes for detecting synergistic effects (e.g., via isobologram or Chou-Talalay method). Randomization, blinding, and strict stopping criteria for in vivo toxicity are not addressed.",

"flags": [

"generic_template",

"missing_synergy_analysis_plan",

"missing_power_rationale",
"missing_randomization_blinding"
]
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}

Ethics Safety Governance

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"reviews": [
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"candidate_id": "candidate_c8927343aa144a7f",
"review_type": "impact_safety",
"score": 4,
"feedback": "The direct testing of combination therapies poses standard oncological research risks. Institutional Animal Care and Use Committee (IACUC) approval is mandatory for all in vivo murine models. If utilizing primary human patient-derived xenografts (PDX) or clinical samples, Institutional Review Board (IRB) approval and documented informed consent are required. Biosafety Level 2 (BSL-2) protocols must be maintained for handling human cell lines. Dual-use risk is low. The primary safety constraint is monitoring for overlapping dose-limiting toxicities; the prespecified stopping criterion of >20% weight loss or unacceptable distress in animals must be strictly enforced.",
"fatal_flaws": []
},
{
"candidate_id": "candidate_7216a3e5cc304123",
"review_type": "impact_safety",
"score": 4,
"feedback": "Testing for observable intermediates (mediation) often requires serial sampling or additional invasive procedures in animal models. The IACUC protocol must explicitly justify the frequency and volume of blood draws or serial biopsies to ensure animal welfare (adherence to the 3Rs: Replacement, Reduction, Refinement). IRB approval is required for any human-derived tissue assays. Data privacy standards must be upheld if correlating intermediates with patient clinical data.",
"fatal_flaws": []
},
{
"candidate_id": "candidate_af60aaba2ec748e4",
"review_type": "impact_safety",
"score": 4,
"feedback": "Testing boundary conditions may involve dose-escalation or extreme physiological states to force mechanism failure. This increases the risk of severe adverse events in vivo. The experimental design must include rigid, predefined humane endpoints and safety limits (e.g., maximum tolerated dose thresholds, <20% weight loss) to prevent unnecessary animal suffering. Standard BSL-2 and IACUC/IRB governance applies.",
"fatal_flaws": []
},
{
"candidate_id": "candidate_a44b947cc6be4874",
"review_type": "impact_safety",
"score": 4,
"feedback": "Head-to-head comparisons of multiple competing mechanisms and combinations will require multiple experimental arms, significantly increasing the required animal sample size. The investigator must provide a robust statistical power rationale to the IACUC to justify the number of animals used, ensuring compliance with the 'Reduction' principle of the 3Rs. All human data and PDX models require IRB oversight and de-identification.",
"fatal_flaws": []
},
{
"candidate_id": "candidate_8f7cc51dc8214718",

"review_type": "impact_safety",

"score": 4,

"feedback": "Transferring a mechanism or combination strategy from an adjacent system (e.g., a different cancer type or pathway) carries the risk of unexpected synergistic toxicities in NSCLC models. Initial in vivo experiments must be designed as cautious dose-finding (Phase 0/1 equivalent) studies with intensive monitoring. Full IACUC and IRB approvals are required, and any genomic data generated must comply with institutional data rights and sharing policies.",

"fatal_flaws": []

},

{

"candidate_id": "candidate_c1d9726261244b0c",

"review_type": "impact_safety",

"score": 4,

"feedback": "Evaluating prespecified non-additive (synergistic) interactions inherently risks synergistic toxicity. The protocol must explicitly define independent and dependent variables for toxicity (e.g., liver enzymes, weight loss, behavioral changes) alongside efficacy. Strict adherence to the >20% weight loss stopping criterion is critical. BSL-2 safety for human cell lines and IACUC/IRB approvals for animal and human tissues, respectively, are non-negotiable prerequisites.",

"fatal_flaws": []

}

]

}

Impact Review

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"reviews": [

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"candidate_id": "candidate_c8927343aa144a7f",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test the most directly supported intervention...') and proposes no specific KRAS inhibitor, secondary agent, or resistance mechanism.",

"importance": "None. Lacks any specific biological or clinical hypothesis regarding NSCLC.",

"feasibility": "Unassessable. No specific experimental models, drugs, or assays are proposed.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided to evaluate generalizability."

},

{

"candidate_id": "candidate_7216a3e5cc304123",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test whether the proposed effect is mediated by an observable intermediate...') and fails to specify the intermediate, the KRAS inhibitor, or the resistance context.",

"importance": "None. Lacks a concrete biological mechanism (e.g., genomic, cell-state, or TME) to evaluate.",

"feasibility": "Unassessable. No specific mediator or assay is defined.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_af60aaba2ec748e4",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test a boundary condition...') and does not define the primary mechanism or the boundary condition itself.",

"importance": "None. Cannot be determined without a specific KRAS resistance mechanism.",

"feasibility": "Unassessable. Lacks specific experimental design.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.", "time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_a44b947cc6be4874",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Compare the primary mechanism with the strongest competing explanation...') and does not identify the primary mechanism or the competing explanation.",

"importance": "None. No specific scientific claim is made regarding NSCLC KRAS resistance.",

"feasibility": "Unassessable. No specific comparators are provided.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_8f7cc51dc8214718",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Transfer a validated mechanism from an adjacent system...') and does not specify the mechanism or the system.",

"importance": "None. Lacks a concrete biological hypothesis.",

"feasibility": "Unassessable. No specific mechanism is proposed for transfer.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

},

{

"candidate_id": "candidate_c1d9726261244b0c",

"expected_information_gain": "Zero. The candidate is an uninstantiated template ('Test a combined intervention for a prespecified non-additive interaction...') and fails to specify the drugs or the resistance context.",

"importance": "None. Cannot evaluate clinical relevance without specific agents.",

"feasibility": "Unassessable. No specific combination therapy is defined.",

"cost_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"time_estimate": "Unknown. Cannot be estimated without a defined intervention.",

"external_validity": "Unassessable. No specific context or mechanism is provided."

}

]

}

<details><summary>Validated typed payload</summary>

{}

</details>

`artifact_b05b34f56d8a47f5` — ranking

- Stage: `rank`
- Type: `specialist_output`
- Schema: `TournamentState`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_6729074f7f9d49e4`, `artifact_9b8d2c50cede41de`, `artifact_6abc279774ef4fdc`, `artifact_5cf5fdcceee84d46`, `artifact_5a95d55e05e54405`, `artifact_aa9f0172d103422d`

- Checksum: `e495c19cf50371ec49d79de20fa6d05da8348af4ee27f9a0dd07a4abd2469b65`

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      "final_elo": 1290,
      "score_movement": 90,
      "rank": 1
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      "rank": 2
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      "initial_elo": 1200,
      "final_elo": 1203,
      "score_movement": 3,
      "rank": 3
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      "id": "candidate_7216a3e5cc304123",
      "initial_elo": 1200,
      "final_elo": 1184,
      "score_movement": -16,
      "rank": 4
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      "initial_elo": 1200,
      "final_elo": 1184,
      "score_movement": -16,
      "rank": 5
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      "initial_elo": 1200,
      "final_elo": 1184,
      "score_movement": -16,
      "rank": 6
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      "initial_elo": 1200,

      "final_elo": 1171,

      "score_movement": -29,

      "rank": 7

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      "id": "candidate_8b7f18e851d94646",

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      "final_elo": 1152,

      "score_movement": -48,

      "rank": 8

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  "swiss_rounds": [

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          "winner": "candidate_c8927343aa144a7f",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

        },

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          "candidate_a": "candidate_af60aaba2ec748e4",

          "candidate_b": "candidate_a44b947cc6be4874",

          "winner": "candidate_af60aaba2ec748e4",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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          "candidate_a": "candidate_8f7cc51dc8214718",

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          "winner": "candidate_8f7cc51dc8214718",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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          "winner": "candidate_01977dd445254834",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

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          "winner": "candidate_c8927343aa144a7f",

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          "elo_change_b": -16,

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          "winner": "candidate_7216a3e5cc304123",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

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          "candidate_a": "candidate_c1d9726261244b0c",

          "candidate_b": "candidate_8b7f18e851d94646",

          "winner": "candidate_c1d9726261244b0c",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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          "candidate_b": "candidate_8f7cc51dc8214718",

          "winner": "candidate_c8927343aa144a7f",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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        {

          "candidate_a": "candidate_af60aaba2ec748e4",

          "candidate_b": "candidate_7216a3e5cc304123",

          "winner": "candidate_af60aaba2ec748e4",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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          "candidate_a": "candidate_01977dd445254834",

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          "winner": "candidate_01977dd445254834",

          "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

          "elo_change_a": 16,

          "elo_change_b": -16,

          "confidence": 0.1

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          "winner": "candidate_a44b947cc6be4874",
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          "elo_change_b": -16,
          "confidence": 0.1
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      "candidate_8f7cc51dc8214718",
      "candidate_af60aaba2ec748e4",
      "candidate_01977dd445254834"
    ],
    "comparisons": [
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        "winner": "candidate_c8927343aa144a7f",
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        "winner": "candidate_af60aaba2ec748e4",
        "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",
        "elo_change_a": 16,
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        "winner": "candidate_c8927343aa144a7f",
        "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",
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        "candidate_b": "candidate_01977dd445254834",

        "winner": "candidate_8f7cc51dc8214718",

        "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

        "elo_change_a": 16,

        "elo_change_b": -16,

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        "candidate_b": "candidate_01977dd445254834",

        "winner": "candidate_c8927343aa144a7f",

        "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

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        "candidate_a": "candidate_8f7cc51dc8214718",

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        "winner": "candidate_8f7cc51dc8214718",

        "rationale": "Both candidates are generic boilerplate templates lacking specific hypotheses. Candidate A is selected marginally based on structural completeness of the template.",

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        "elo_change_b": -15,

        "confidence": 0.1

      }

    ]

  },

  "shortlist": [

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    "candidate_8f7cc51dc8214718",

    "candidate_af60aaba2ec748e4",

    "candidate_01977dd445254834"

  ],

  "convergence_state": "unstable",

  "uncertainty_statement": "All candidates generated are generic boilerplate templates lacking specific, testable scientific hypotheses regarding KRAS inhibitor resistance. No verified evidence claims support these candidates. The ranking is structurally complete but scientifically arbitrary due to the fatal flaws identified in the reflection phase."

}

<details><summary>Validated typed payload</summary>

{

  "ratings": {
    "candidate_01977dd445254834": 1562.5304984710244,
    "candidate_7216a3e5cc304123": 1546.597626351594,
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    "candidate_a44b947cc6be4874": 1500.0,
    "candidate_af60aaba2ec748e4": 1486.8718751773815,
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        "candidate_7216a3e5cc304123": 1500.0
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      "candidate_b_id": "candidate_7216a3e5cc304123",
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        "candidate_7216a3e5cc304123": 3.5
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      "presented_first_id": "candidate_01977dd445254834"
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      "id": "comparison_a971dd6ee36f43d4",
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      "rationale": "Compared evidence status, validity, novelty, feasibility, impact, risk, reproducibility, and information gain.",
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        "candidate_8f7cc51dc8214718": 1500.0

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      "confidence": 0.55,

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      "confidence": 0.55,

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      "confidence": 0.55,

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        "candidate_8b7f18e851d94646": 1516.0

      },

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      "rubric_version": "1",

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      "confidence": 0.55,

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        "candidate_c1d9726261244b0c": 3.5

      },

      "presented_first_id": "candidate_c1d9726261244b0c"

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      "confidence": 0.55,

      "elo_before": {

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      "candidate_b_id": "candidate_c8927343aa144a7f",

      "rubric_version": "1",

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</details>

`artifact_84ba9dcbe2764e4f` — supervisor

- Stage: `rank`
- Type: `stage_bundle`
- Schema: `markdown`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_b05b34f56d8a47f5`
- Checksum: `58fa52efc0f453deaa82515e058003be941c6c2f48d248d56ea8dbcaa5441b2e`

Ranking

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          "elo_change_b": -16,

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    "candidate_01977dd445254834"

  ],

  "convergence_state": "unstable",

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}

<details><summary>Validated typed payload</summary>

{}

</details>

`artifact_afd01fb2c4e04e9d` — evolution

- Stage: `evolve`
- Type: `specialist_output`
- Schema: `EvolutionCycle`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_6abc279774ef4fdc`, `artifact_5cf5fdcceee84d46`, `artifact_5a95d55e05e54405`, `artifact_aa9f0172d103422d`, `artifact_b05b34f56d8a47f5`, `artifact_84ba9dcbe2764e4f`
- Checksum: `c5dd7235f06594a05e2bedcfd9de6061ebb39a526a22f661c5e52d7d277fa8e2`

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          "claim": "Sequential administration of a PD-L1 inhibitor following acquired resistance to sotorasib (KRAS G12C inhibitor) in NSCLC models overcomes resistance driven by the JAK2/STAT3/IL-6 pathway-mediated immunosuppressive tumor microenvironment.",

          "protocol": "1. Establish syngeneic KRAS G12C mutant NSCLC mouse models (e.g., LLC-KrasG12C) under IACUC approval and BSL-2 compliance. 2. Treat mice with sotorasib until acquired resistance develops (defined by tumor regrowth). 3. Randomize resistant mice to four groups: vehicle, continued sotorasib monotherapy, PD-L1 inhibitor monotherapy, or sotorasib + PD-L1 inhibitor combination (independent variables). 4. Measure tumor volume over time (dependent variable) with sample size powered to detect a p < 0.05 difference. 5. Perform flow cytometry and IHC on tumor tissues to quantify CD8+ T cells and MDSCs, and assess JAK2/STAT3/IL-6 pathway activation to validate the TME mechanism.",

          "falsifier": "The combination of sotorasib and PD-L1 inhibitor fails to significantly reduce tumor burden (p < 0.05) compared to sotorasib monotherapy in resistant models, fails to restore CD8+ T cell infiltration, or causes unacceptable toxicity (>20% body weight loss).",

          "assumptions": [

            "Syngeneic mouse models accurately reflect human TME and immune responses.",

            "Resistance is driven by the specified immune evasion mechanism."

          ],

          "limitations": [

            "Preclinical models may not fully capture human immune system complexities.",

            "Optimal sequencing in humans may differ."

          ]

        },

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          "id": "candidate_8f7cc51dc8214718_v2",

          "parent_id": "candidate_8f7cc51dc8214718",

          "claim": "Dual inhibition of mTORC1/2 (sapanisertib) combined with a KRAS G12C inhibitor (sotorasib) overcomes acquired resistance in NSCLC models lacking secondary KRAS mutations, driven by PI3K/AKT/mTOR pathway activation.",

          "protocol": "1. Generate sotorasib-resistant KRAS G12C NSCLC cell lines and patient-derived xenograft (PDX) models lacking secondary KRAS mutations (IRB and IACUC approved). 2. In vitro: Treat resistant cells with varying concentrations of sotorasib, sapanisertib, or the combination (independent variables); assess cell viability (dependent variable) and calculate Combination Index (CI). 3. In vivo: Randomize PDX mice to vehicle, sotorasib, sapanisertib, or combination. 4. Measure tumor volume and monitor body weight for toxicity. 5. Perform Western blot on tumor lysates for p4E-BP1 and pAKT to confirm target engagement and validate the cell-state mechanism.",

          "falsifier": "The combination fails to achieve synergistic efficacy (CI < 1) in vitro, fails to significantly reduce tumor volume in vivo (p < 0.05) compared to monotherapy, fails to inhibit p4E-BP1/pAKT, or causes >20% body weight loss.",

          "assumptions": [

            "Resistance in these models is primarily driven by PI3K/AKT/mTOR activation.",

            "The combination can be administered at tolerable doses without overlapping dose-limiting toxicities."

          ],

          "limitations": [

            "Toxicity of dual mTORC1/2 inhibition combined with KRAS inhibition may be dose-limiting in humans."

          ]

        },

        {

          "id": "candidate_af60aaba2ec748e4_v2",

          "parent_id": "candidate_af60aaba2ec748e4",

          "claim": "Combining a YAP-TEAD inhibitor with a KRAS G12C inhibitor (sotorasib) overcomes cell-state plasticity (EMT) mediated resistance in KRAS-mutant NSCLC.",

          "protocol": "1. Identify or generate KRAS G12C NSCLC cell lines exhibiting EMT-mediated resistance to sotorasib. 2. Treat with varying concentrations of sotorasib, a YAP-TEAD inhibitor, or the combination (independent variables). 3. Assess cell viability in vitro (calculate CI) and tumor growth in vivo using xenograft models under IACUC approval (dependent variables). 4. Measure EMT markers (E-cadherin, Vimentin) and cancer stemness markers via RT-qPCR and Western blot to validate the cell-state mechanism.",

          "falsifier": "The combination does not show synergistic efficacy (CI > 1) in vitro, fails to significantly reduce tumor burden in vivo (p < 0.05), fails to reverse the EMT signature in resistant models, or causes >20% weight loss.",

          "assumptions": [

            "YAP1/TEAD activation is a primary driver of EMT in these resistant models."

          ],

          "limitations": [

            "EMT is a complex process; other pathways may compensate if YAP-TEAD is inhibited."

          ]

        },

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          "id": "candidate_7216a3e5cc304123_v2",

          "parent_id": "candidate_7216a3e5cc304123",

          "claim": "In KRAS G12C NSCLC with acquired Y96D secondary mutations, a pan-KRAS inhibitor overcomes resistance that is refractory to sotorasib and adagrasib.",

          "protocol": "1. Engineer NSCLC cell lines expressing KRAS G12C/Y96D double mutations. 2. Treat with sotorasib, adagrasib, and a selected pan-KRAS inhibitor (independent variables). 3. Measure cell viability (calculate CI) and KRAS downstream signaling (pERK, pMEK) as dependent variables. 4. Establish xenograft models with the double mutant (IACUC approved) and assess in vivo tumor growth inhibition by the pan-KRAS agent compared to vehicle.",

          "falsifier": "The pan-KRAS inhibitor fails to inhibit cell viability, fails to reduce pERK signaling, fails to significantly reduce tumor volume (p < 0.05) in the G12C/Y96D double mutant models, or causes >20% weight loss.",

          "assumptions": [

            "The Y96D mutation confers resistance to both sotorasib and adagrasib but remains susceptible to pan-KRAS inhibition."

          ],

          "limitations": [

            "Requires access to specific novel pan-KRAS inhibitors which may have different pharmacokinetic profiles."

          ]

        }

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          "reviewer": "evolution",

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          "protocol": "1. Establish syngeneic KRAS G12C mutant NSCLC mouse models (e.g., LLC-KrasG12C) under IACUC approval and BSL-2 compliance. 2. Treat mice with sotorasib until acquired resistance develops (defined by tumor regrowth). 3. Randomize resistant mice to four groups: vehicle, continued sotorasib monotherapy, PD-L1 inhibitor monotherapy, or sotorasib + PD-L1 inhibitor combination (independent variables). 4. Measure tumor volume over time (dependent variable) with sample size powered to detect a p < 0.05 difference. 5. Perform flow cytometry and IHC on tumor tissues to quantify CD8+ T cells and MDSCs, and assess JAK2/STAT3/IL-6 pathway activation to validate the TME mechanism.",

          "falsifier": "The combination of sotorasib and PD-L1 inhibitor fails to significantly reduce tumor burden (p < 0.05) compared to sotorasib monotherapy in resistant models, fails to restore CD8+ T cell infiltration, or causes unacceptable toxicity (>20% body weight loss).",

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            "Syngeneic mouse models accurately reflect human TME and immune responses.",

            "Resistance is driven by the specified immune evasion mechanism."

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          "limitations": [

            "Preclinical models may not fully capture human immune system complexities.",

            "Optimal sequencing in humans may differ."

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          "protocol": "1. Generate sotorasib-resistant KRAS G12C NSCLC cell lines and patient-derived xenograft (PDX) models lacking secondary KRAS mutations (IRB and IACUC approved). 2. In vitro: Treat resistant cells with varying concentrations of sotorasib, sapanisertib, or the combination (independent variables); assess cell viability (dependent variable) and calculate Combination Index (CI). 3. In vivo: Randomize PDX mice to vehicle, sotorasib, sapanisertib, or combination. 4. Measure tumor volume and monitor body weight for toxicity. 5. Perform Western blot on tumor lysates for p4E-BP1 and pAKT to confirm target engagement and validate the cell-state mechanism.",

          "falsifier": "The combination fails to achieve synergistic efficacy (CI < 1) in vitro, fails to significantly reduce tumor volume in vivo (p < 0.05) compared to monotherapy, fails to inhibit p4E-BP1/pAKT, or causes >20% body weight loss.",

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            "Resistance in these models is primarily driven by PI3K/AKT/mTOR activation.",

            "The combination can be administered at tolerable doses without overlapping dose-limiting toxicities."

          ],

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          "protocol": "1. Identify or generate KRAS G12C NSCLC cell lines exhibiting EMT-mediated resistance to sotorasib. 2. Treat with varying concentrations of sotorasib, a YAP-TEAD inhibitor, or the combination (independent variables). 3. Assess cell viability in vitro (calculate CI) and tumor growth in vivo using xenograft models under IACUC

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"The proposed mediator or discriminating measurement changes first.",
"The primary result survives the prespecified robustness analysis in evolution round 1.",
"The primary result survives the prespecified robustness analysis in evolution round 2.",
"The primary result survives the prespecified robustness analysis in evolution round 3."
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"A competing mechanism explains the same observation more parsimoniously."
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"Unspecified effect size and external-validity assumptions."
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"Added a robustness prediction and explicit re-review requirement."
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"Bias, confounding, or an adverse safety signal may invalidate promotion."
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"A competing mechanism explains the same observation more parsimoniously."
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"Promotion remains conditional on verified evidence and execution of the proposed test."
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"Promotion remains conditional on verified evidence and execution of the proposed test."
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- Parent: `none`
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"claim": "Sequential administration of a PD-L1 inhibitor following acquired resistance to sotorasib (KRAS G12C inhibitor) in NSCLC models overcomes resistance driven by the JAK2/STAT3/IL-6 pathway-mediated immunosuppressive tumor microenvironment.",

"protocol": "1. Establish syngeneic KRAS G12C mutant NSCLC mouse models (e.g., LLC-KrasG12C) under IACUC approval and BSL-2 compliance. 2. Treat mice with sotorasib until acquired resistance develops (defined by tumor regrowth). 3. Randomize resistant mice to four groups: vehicle, continued sotorasib monotherapy, PD-L1 inhibitor monotherapy, or sotorasib + PD-L1 inhibitor combination (independent variables). 4. Measure tumor volume over time (dependent variable) with sample size powered to detect a p < 0.05 difference. 5. Perform flow cytometry and IHC on tumor tissues to quantify CD8+ T cells and MDSCs, and assess JAK2/STAT3/IL-6 pathway activation to validate the TME mechanism.",

"falsifier": "The combination of sotorasib and PD-L1 inhibitor fails to significantly reduce tumor burden (p < 0.05) compared to sotorasib monotherapy in resistant models, fails to restore CD8+ T cell infiltration, or causes unacceptable toxicity (>20% body weight loss).",

"assumptions": [

"Syngeneic mouse models accurately reflect human TME and immune responses.",

"Resistance is driven by the specified immune evasion mechanism."

],

"limitations": [

"Preclinical models may not fully capture human immune system complexities.",

"Optimal sequencing in humans may differ."

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"claim": "Dual inhibition of mTORC1/2 (sapanisertib) combined with a KRAS G12C inhibitor (sotorasib) overcomes acquired resistance in NSCLC models lacking secondary KRAS mutations, driven by PI3K/AKT/mTOR pathway activation.",

"protocol": "1. Generate sotorasib-resistant KRAS G12C NSCLC cell lines and patient-derived xenograft (PDX) models lacking secondary KRAS mutations (IRB and IACUC approved). 2. In vitro: Treat resistant cells with varying concentrations of sotorasib, sapanisertib, or the combination (independent variables); assess cell viability (dependent variable) and calculate Combination Index (CI). 3. In vivo: Randomize PDX mice to vehicle, sotorasib, sapanisertib, or combination. 4. Measure tumor volume and monitor body weight for toxicity. 5. Perform Western blot on tumor lysates for p4E-BP1 and pAKT to confirm target engagement and validate the cell-state mechanism.",

"falsifier": "The combination fails to achieve synergistic efficacy (CI < 1) in vitro, fails to significantly reduce tumor volume in vivo (p < 0.05) compared to monotherapy, fails to inhibit p4E-BP1/pAKT, or causes >20% body weight loss.",

"assumptions": [

"Resistance in these models is primarily driven by PI3K/AKT/mTOR activation.",

"The combination can be administered at tolerable doses without overlapping dose-limiting toxicities."

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"limitations": [

"Toxicity of dual mTORC1/2 inhibition combined with KRAS inhibition may be dose-limiting in humans."

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"protocol": "1. Identify or generate KRAS G12C NSCLC cell lines exhibiting EMT-mediated resistance to sotorasib. 2. Treat with varying concentrations of sotorasib, a YAP-TEAD inhibitor, or the combination (independent variables). 3. Assess cell viability in vitro (calculate CI) and tumor growth in vivo using xenograft models under IACUC approval (dependent variables). 4. Measure EMT markers (E-cadherin, Vimentin) and cancer stemness markers via RT-qPCR and Western blot to validate the cell-state mechanism.",

"falsifier": "The combination does not show synergistic efficacy (CI > 1) in vitro, fails to significantly reduce tumor burden in vivo (p < 0.05), fails to reverse the EMT signature in resistant models, or causes >20% weight loss.",

"assumptions": [

"YAP1/TEAD activation is a primary driver of EMT in these resistant models."

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"limitations": [

"EMT is a complex process; other pathways may compensate if YAP-TEAD is inhibited."

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"protocol": "1. Engineer NSCLC cell lines expressing KRAS G12C/Y96D double mutations. 2. Treat with sotorasib, adagrasib, and a selected pan-KRAS inhibitor (independent variables). 3. Measure cell viability (calculate CI) and KRAS downstream signaling (pERK, pMEK) as dependent variables. 4. Establish xenograft models with the double mutant (IACUC approved) and assess in vivo tumor growth inhibition by the pan-KRAS agent compared to vehicle.",

"falsifier": "The pan-KRAS inhibitor fails to inhibit cell viability, fails to reduce pERK signaling, fails to significantly reduce tumor volume (p < 0.05) in the G12C/Y96D double mutant models, or causes >20% weight loss.",

"assumptions": [

"The Y96D mutation confers resistance to both sotorasib and adagrasib but remains susceptible to pan-KRAS inhibition."

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"Requires access to specific novel pan-KRAS inhibitors which may have different pharmacokinetic profiles."

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"score": 88,

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"score": 92,

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"feedback": "Addresses genomic resistance (Y96D) effectively, but relies on generalized pan-KRAS inhibitors rather than a specific combination strategy."

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"protocol": "1. Establish syngeneic KRAS G12C mutant NSCLC mouse models (e.g., LLC-KrasG12C) under IACUC approval and BSL-2 compliance. 2. Treat mice with sotorasib until acquired resistance develops (defined by tumor regrowth). 3. Randomize resistant mice to four groups: vehicle, continued sotorasib monotherapy, PD-L1 inhibitor monotherapy, or sotorasib + PD-L1 inhibitor combination (independent variables). 4. Measure tumor volume over time (dependent variable) with sample size powered to detect a p < 0.05 difference. 5. Perform flow cytometry and IHC on tumor tissues to quantify CD8+ T cells and MDSCs, and assess JAK2/STAT3/IL-6 pathway activation to validate the TME mechanism.",

"falsifier": "The combination of sotorasib and PD-L1 inhibitor fails to significantly reduce tumor burden (p < 0.05) compared to sotorasib monotherapy in resistant models, fails to restore CD8+ T cell infiltration, or causes unacceptable toxicity (>20% body weight loss).",

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"Syngeneic mouse models accurately reflect human TME and immune responses.",

"Resistance is driven by the specified immune evasion mechanism."

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"limitations": [

"Preclinical models may not fully capture human immune system complexities.",

"Optimal sequencing in humans may differ."

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"protocol": "1. Generate sotorasib-resistant KRAS G12C NSCLC cell lines and patient-derived xenograft (PDX) models lacking secondary KRAS mutations (IRB and IACUC approved). 2. In vitro: Treat resistant cells with varying concentrations of sotorasib, sapanisertib, or the combination (independent variables); assess cell viability (dependent variable) and calculate Combination Index (CI). 3. In vivo: Randomize PDX mice to vehicle, sotorasib, sapanisertib, or combination. 4. Measure tumor volume and monitor body weight for toxicity. 5. Perform Western blot on tumor lysates for p4E-BP1 and pAKT to confirm target engagement and validate the cell-state mechanism.",

"falsifier": "The combination fails to achieve synergistic efficacy (CI < 1) in vitro, fails to significantly reduce tumor volume in vivo (p < 0.05) compared to monotherapy, fails to inhibit p4E-BP1/pAKT, or causes >20% body weight loss.",

"assumptions": [

"Resistance in these models is primarily driven by PI3K/AKT/mTOR activation.",

"The combination can be administered at tolerable doses without overlapping dose-limiting toxicities."

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"limitations": [

"Toxicity of dual mTORC1/2 inhibition combined with KRAS inhibition may be dose-limiting in humans."

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"protocol": "1. Identify or generate KRAS G12C NSCLC cell lines exhibiting EMT-mediated resistance to sotorasib. 2. Treat with varying concentrations of sotorasib, a YAP-TEAD inhibitor, or the combination (independent variables). 3. Assess cell viability in vitro (calculate CI) and tumor growth in vivo using xenograft models under IACUC

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- Type: `specialist_output`
- Schema: `ResearchLandscape`
- Status: `accepted`
- Version: 1
- Parent: `none`
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"outcome": "Acquired resistance to KRAS G12C inhibitors (e.g., sotorasib) overcome by sequential PD-L1 inhibition.",

"evidence_overlap": "Supported by unverified preclinical syngeneic mouse model leads (src_e081381f915c4d0c).",

"data_needs": "Clinical translation data, optimal sequencing schedules to avoid overlapping toxicities, and verified source data."

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"name": "Cell-State Plasticity and Bypass Signaling",

"mechanism": "Activation of PI3K/AKT/mTOR pathway (p4E-BP1 upregulation) and YAP1/TEAD-driven epithelial-to-mesenchymal transition (EMT) and stemness.",

"outcome": "Restoration of sensitivity via PI3K/mTORC1/2 dual inhibition (copanlisib/sapanisertib) or YAP-TEAD inhibitors.",

"evidence_overlap": "Supported by unverified in vitro and PDX/CDX model leads lacking secondary genomic alterations (src_a800a0097e02472e, src_6022e896f7e74468).",

"data_needs": "In vivo tolerability data for dual targeted combinations and specific transcriptomic datasets (with accession numbers) to validate EMT/stemness states."

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"name": "Genomic Alterations",

"mechanism": "Secondary KRAS mutations (e.g., Y96D) altering drug binding and mutations in RAS-MAPK pathway genes.",

"outcome": "Reactivation of signaling leading to resistance to adagrasib/sotorasib.",

"evidence_overlap": "Supported by unverified clinical case reports and small patient cohorts (src_1a2978ba8af34cb0).",

"data_needs": "Large-cohort genomic profiling to determine mutation frequencies and structural data for next-generation inhibitor design."

}

],

"contradictions": [

{

"description": "Targeted vs. Broad Resistance Mechanisms",

"conflict": "Genomic resistance mechanisms (e.g., Y96D) are highly heterogeneous and patient-specific, complicating a single combination strategy. In contrast, cell-state (EMT) and TME (immunosuppression) mechanisms suggest broader, potentially universal bypass routes that could be targeted with a uniform combination strategy (e.g., PD-L1 or mTOR inhibitors)."

}

],

"verified_evidence": [],

"unverified_leads": [

{

"id": "src_e081381f915c4d0c",

"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive TME via JAK2/STAT3/IL-6, mediating resistance to sotorasib. Sequential PD-L1 inhibition re-sensitizes tumors.",

"source": "J Immunother Cancer. 2026 Feb 4;14(2):e012886."

},

{

"id": "src_a800a0097e02472e",

"claim": "Acquired resistance without secondary genomic alterations is driven by PI3K/AKT/mTOR pathway. PI3K or mTORC1/2 dual inhibition restores sensitivity.",

"source": "eLife / bioRxiv (2025)"

},

{

"id": "src_1a2978ba8af34cb0",

"claim": "Acquired resistance involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) and RAS-MAPK pathway genes.",

"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al."

},

{

"id": "src_6022e896f7e74468",

"claim": "YAP1/TEAD pathway drives resistance by promoting EMT and cancer stemness. Combining KRAS G12C inhibitors with YAP-TEAD inhibitors shows synergy.",

"source": "Frontiers in Oncology / AACR Annual Meeting 2025"

}

],

"data_needs": [

"Verification of all discovered literature leads to convert them into verified evidence.",
"Specific datasets or accession numbers for any single-cell or spatial transcriptomics claims regarding TME or cell-state plasticity.",
"In vivo safety and toxicity data for combining KRAS inhibitors with PI3K/mTOR or YAP-TEAD inhibitors to rule out overlapping dose-limiting toxicities.",
"Discriminating assays to isolate whether resistance in a given model is primarily genomic, cell-state, or TME-driven."
]
}

<details><summary>Validated typed payload</summary>

{
  "clusters": [
    {
      "id": "cluster_12ae4986e64e4ee5",
      "name": "Evidence First",
      "candidate_ids": [
        "candidate_c8927343aa144a7f",
        "candidate_7216a3e5cc304123"
      ],
      "required_data": [
        "Verified evidence",
        "Mode-appropriate measurements"
      ],
      "schema_version": "1.0",
      "shared_outcome": "The prespecified research outcome.",
      "evidence_overlap": [],
      "shared_mechanism": "Candidates share a generation lens but retain distinct predictions."
    },
    {
      "id": "cluster_6dc2570a39304d6f",
      "name": "Mechanism First",
      "candidate_ids": [
        "candidate_af60aaba2ec748e4",
        "candidate_a44b947cc6be4874"
      ],
      "required_data": [
        "Verified evidence",
        "Mode-appropriate measurements"
      ],
      "schema_version": "1.0",
      "shared_outcome": "The prespecified research outcome.",
      "evidence_overlap": [],
      "shared_mechanism": "Candidates share a generation lens but retain distinct predictions."
    },
    {
      "id": "cluster_9cdf33bd99fc4d2e",
      "name": "Analogy Transfer",
      "candidate_ids": [
        "candidate_8f7cc51dc8214718",

        "candidate_c1d9726261244b0c"

      ],

      "required_data": [

        "Verified evidence",

        "Mode-appropriate measurements"

      ],

      "schema_version": "1.0",

      "shared_outcome": "The prespecified research outcome.",

      "evidence_overlap": [],

      "shared_mechanism": "Candidates share a generation lens but retain distinct predictions."

    },

    {

      "id": "cluster_47560164db174407",

      "name": "Competing Explanation",

      "candidate_ids": [

        "candidate_01977dd445254834",

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      ],

      "schema_version": "1.0",

      "shared_outcome": "The prespecified research outcome.",

      "evidence_overlap": [],

      "shared_mechanism": "Candidates share a generation lens but retain distinct predictions."

    }

  ],

  "duplicates": [],

  "coverage_gaps": [

    "Independent replication evidence",

    "Negative and null-result evidence",

    "External-validity boundary conditions"

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    "candidate_01977dd445254834"

  ]

}

</details>

`artifact_daedc3314bcc483b` — supervisor

- Stage: `proximity`
- Type: `stage_bundle`
- Schema: `markdown`
- Status: `accepted`

- Version: 1
- Parent: `none`
- Inputs: `artifact_623f0b27987f43a4`
- Checksum: `01b186ee15f9e36dacbae2cdffeb585706c5f4863c1d7ea6f22daf092b85333c`

Proximity

{
"schema_version": "1.0",
"clusters": [
{
"name": "Tumor-Microenvironment (TME) Immune Evasion",
"mechanism": "JAK2/STAT3/IL-6 pathway activation leading to PD-L1 upregulation, CD8+ T cell reduction, and MDSC expansion.",
"outcome": "Acquired resistance to KRAS G12C inhibitors (e.g., sotorasib) overcome by sequential PD-L1 inhibition.",
"evidence_overlap": "Supported by unverified preclinical syngeneic mouse model leads (src_e081381f915c4d0c).",
"data_needs": "Clinical translation data, optimal sequencing schedules to avoid overlapping toxicities, and verified source data."
},
{
"name": "Cell-State Plasticity and Bypass Signaling",
"mechanism": "Activation of PI3K/AKT/mTOR pathway (p4E-BP1 upregulation) and YAP1/TEAD-driven epithelial-to-mesenchymal transition (EMT) and stemness.",
"outcome": "Restoration of sensitivity via PI3K/mTORC1/2 dual inhibition (copanlisib/sapanisertib) or YAP-TEAD inhibitors.",
"evidence_overlap": "Supported by unverified in vitro and PDX/CDX model leads lacking secondary genomic alterations (src_a800a0097e02472e, src_6022e896f7e74468).",
"data_needs": "In vivo tolerability data for dual targeted combinations and specific transcriptomic datasets (with accession numbers) to validate EMT/stemness states."
},
{
"name": "Genomic Alterations",
"mechanism": "Secondary KRAS mutations (e.g., Y96D) altering drug binding and mutations in RAS-MAPK pathway genes.",
"outcome": "Reactivation of signaling leading to resistance to adagrasib/sotorasib.",
"evidence_overlap": "Supported by unverified clinical case reports and small patient cohorts (src_1a2978ba8af34cb0).",
"data_needs": "Large-cohort genomic profiling to determine mutation frequencies and structural data for next-generation inhibitor design."
}
],
"contradictions": [
{
"description": "Targeted vs. Broad Resistance Mechanisms",
"conflict": "Genomic resistance mechanisms (e.g., Y96D) are highly heterogeneous and patient-specific, complicating a single combination strategy. In contrast, cell-state (EMT) and TME (immunosuppression) mechanisms suggest broader, potentially universal bypass routes that could be targeted with a uniform combination strategy (e.g., PD-L1 or mTOR inhibitors)."
}
],
"verified_evidence": [],
"unverified_leads": [
{
"id": "src_e081381f915c4d0c",
"claim": "Upregulation of PD-L1 in KRAS G12C tumors drives an immunosuppressive TME via JAK2/STAT3/IL-6, mediating resistance to sotorasib. Sequential PD-L1 inhibition re-sensitizes tumors.",
"source": "J Immunother Cancer. 2026 Feb 4;14(2):e012886."
},
{

"id": "src_a800a0097e02472e",
"claim": "Acquired resistance without secondary genomic alterations is driven by PI3K/AKT/mTOR pathway. PI3K or mTORC1/2 dual inhibition restores sensitivity.",
"source": "eLife / bioRxiv (2025)"
},
{
"id": "src_1a2978ba8af34cb0",
"claim": "Acquired resistance involves diverse genomic alterations, including secondary KRAS mutations (e.g., Y96D) and RAS-MAPK pathway genes.",
"source": "Cancer Discovery (2022) / NEJM (2021) Awad et al."
},
{
"id": "src_6022e896f7e74468",
"claim": "YAP1/TEAD pathway drives resistance by promoting EMT and cancer stemness. Combining KRAS G12C inhibitors with YAP-TEAD inhibitors shows synergy.",
"source": "Frontiers in Oncology / AACR Annual Meeting 2025"
}
],
"data_needs": [
"Verification of all discovered literature leads to convert them into verified evidence.",
"Specific datasets or accession numbers for any single-cell or spatial transcriptomics claims regarding TME or cell-state plasticity.",
"In vivo safety and toxicity data for combining KRAS inhibitors with PI3K/mTOR or YAP-TEAD inhibitors to rule out overlapping dose-limiting toxicities.",
"Discriminating assays to isolate whether resistance in a given model is primarily genomic, cell-state, or TME-driven."
]
}

<details><summary>Validated typed payload</summary>

{}

</details>

`artifact_2bf0c4a3886a4fed` — meta_reviewer

- Stage: `meta_review`
- Type: `specialist_output`
- Schema: `DossierManifest`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_b05b34f56d8a47f5`, `artifact_84ba9dcbe2764e4f`, `artifact_afd01fb2c4e04e9d`, `artifact_b3494fd1f37942ba`, `artifact_623f0b27987f43a4`, `artifact_daedc3314bcc483b`
- Checksum: `1b09e4be20ebda08518937c466ee4186613ffe16a5e70aa7c93d1f69cbd31d0c`

{
"schema_version": "1.0",
"decision": "NO-GO",
"rationale": "The proposal lacks verified evidence to support any of the candidate hypotheses. All discovered evidence remains in a 'discovered_unverified' state. Furthermore, the initial candidates were generic boilerplate templates, and while evolution introduced specific interventions, they cannot be promoted without verified source material.",
"recommended_candidate_ids": [],
"excluded_candidate_ids": [
"candidate_c8927343aa144a7f",
"candidate_7216a3e5cc304123",

"candidate_af60aaba2ec748e4",

"candidate_a44b947cc6be4874",

"candidate_8f7cc51dc8214718",

"candidate_c1d9726261244b0c",

"candidate_01977dd445254834",

"candidate_8b7f18e851d94646",

"candidate_c8927343aa144a7f_v2",

"candidate_8f7cc51dc8214718_v2"

],

"evidence_requirements": "Verification of the primary sources supporting the proposed mechanisms (e.g., JAK2/STAT3/IL-6 pathway for TME, PI3K/AKT/mTOR for cell-state) is required. Source verification must confirm exact claim support for the specific KRAS inhibitors (e.g., sotorasib) and mutations (e.g., G12C) before any candidate can be recommended.",

"fatal_flaws": [

"No verified evidence claims support any candidate.",

"Initial candidates lacked specific, testable scientific hypotheses and failed to specify exact KRAS inhibitors, mutations, or combination strategies."

]

}

<details><summary>Validated typed payload</summary>

{

  "title": "Co-Scientist Research Dossier: Identify rational combination strategies to overcome primary and acquired resistance to KRAS inhibitors in non-small cell lung cancer. Distinguish genomic, cell-state, and tumor-microenvironment mechanisms and propose discriminating validation experiments.",

  "sections": [

    {

      "key": "executive",

      "title": "Executive synthesis",

      "artifact_ids": [],

      "schema_version": "1.0"

    },

    {

      "key": "scope",

      "title": "Research goal and input sufficiency",

      "artifact_ids": [

        "artifact_ae850f40e063477b",

        "artifact_189ca328a62b4466"

      ],

      "schema_version": "1.0"

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    {

      "key": "evidence",

      "title": "Research directions and evidence ledger",

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    },

    {

      "key": "candidates",

      "title": "Candidate ideas",

      "artifact_ids": [

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        "artifact_23c49dd4582546e6",
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        "artifact_856487f71cef4597"
      ],
      "schema_version": "1.0"
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      "title": "Independent reviews and objections",
      "artifact_ids": [
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        "artifact_9b8d2c50cede41de",
        "artifact_6abc279774ef4fdc",
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        "artifact_5a95d55e05e54405",
        "artifact_aa9f0172d103422d"
      ],
      "schema_version": "1.0"
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      "key": "ranking",
      "title": "Tournament ranking and shortlist",
      "artifact_ids": [
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        "artifact_84ba9dcbe2764e4f"
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    {
      "key": "evolution",
      "title": "Evolution and lineage",
      "artifact_ids": [],
      "schema_version": "1.0"
    },
    {
      "key": "landscape",
      "title": "Research landscape",
      "artifact_ids": [
        "artifact_623f0b27987f43a4",
        "artifact_daedc3314bcc483b"
      ],
      "schema_version": "1.0"
    },
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      "key": "protocol",
      "title": "Validation protocol and go/no-go conditions",
      "artifact_ids": [],

      "schema_version": "1.0"
    },
    {
      "key": "limitations",
      "title": "Limitations, uncertainty, and governance",
      "artifact_ids": [],
      "schema_version": "1.0"
    },
    {
      "key": "appendix",
      "title": "Complete artifact appendix",
      "artifact_ids": [
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        "artifact_189ca328a62b4466",
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        "artifact_23c49dd4582546e6",
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        "artifact_856487f71cef4597",
        "artifact_6729074f7f9d49e4",
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        "artifact_84ba9dcbe2764e4f",
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        "artifact_623f0b27987f43a4",
        "artifact_daedc3314bcc483b"
      ],
      "schema_version": "1.0"
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  "schema_version": "1.0",
  "recommendation_candidate_ids": [
    "candidate_01977dd445254834",
    "candidate_7216a3e5cc304123",
    "candidate_a44b947cc6be4874"
  ],
  "evidence_that_would_change_decision": [
    "Verified primary evidence contradicting the proposed mechanism.",
    "A failed discriminating prediction or unacceptable safety signal.",
    "Independent replication favoring a competing explanation."
  ],
  "unresolved_fatal_flaw_candidate_ids": []
}

</details>

`artifact_9f3842730acf40c7` — supervisor

- Stage: `meta_review`
- Type: `stage_bundle`
- Schema: `markdown`
- Status: `accepted`
- Version: 1
- Parent: `none`
- Inputs: `artifact_2bf0c4a3886a4fed`
- Checksum: `6f1ea235b503469be391c1688d14416786067801fc39b1473933b3cb6cf462f3`

Meta Reviewer

{

"schema_version": "1.0",

"decision": "NO-GO",

"rationale": "The proposal lacks verified evidence to support any of the candidate hypotheses. All discovered evidence remains in a 'discovered_unverified' state. Furthermore, the initial candidates were generic boilerplate templates, and while evolution introduced specific interventions, they cannot be promoted without verified source material.",

"recommended_candidate_ids": [],

"excluded_candidate_ids": [

"candidate_c8927343aa144a7f",

"candidate_7216a3e5cc304123",

"candidate_af60aaba2ec748e4",

"candidate_a44b947cc6be4874",

"candidate_8f7cc51dc8214718",

"candidate_c1d9726261244b0c",

"candidate_01977dd445254834",

"candidate_8b7f18e851d94646",

"candidate_c8927343aa144a7f_v2",

"candidate_8f7cc51dc8214718_v2"

],

"evidence_requirements": "Verification of the primary sources supporting the proposed mechanisms (e.g., JAK2/STAT3/IL-6 pathway for TME, PI3K/AKT/mTOR for cell-state) is required. Source verification must confirm exact claim support for the specific KRAS inhibitors (e.g., sotorasib) and mutations (e.g., G12C) before any candidate can be recommended.",

"fatal_flaws": [

"No verified evidence claims support any candidate.",

"Initial candidates lacked specific, testable scientific hypotheses and failed to specify exact KRAS inhibitors, mutations, or combination strategies."

]

}

<details><summary>Validated typed payload</summary>

{}

</details>

Decision and task audit

Decisions

| Time | Actor | Action | Stage | Artifact | Automatic |

2026-07-28T11:08:12.956388+00:00	auto_approval_policy	accept	scope	`artifact_189ca328a62b4466`	yes
2026-07-28T11:09:02.639979+00:00	auto_approval_policy	accept	generate	`artifact_856487f71cef4597`	yes
2026-07-28T11:09:52.533045+00:00	auto_approval_policy	accept	reflect	`artifact_aa9f0172d103422d`	yes
2026-07-28T11:10:38.445680+00:00	auto_approval_policy	accept	rank	`artifact_84ba9dcbe2764e4f`	yes
2026-07-28T11:11:29.395347+00:00	auto_approval_policy	accept	evolve	`artifact_b3494fd1f37942ba`	yes
2026-07-28T11:11:56.820462+00:00	auto_approval_policy	accept	proximity	`artifact_daedc3314bcc483b`	yes
2026-07-28T11:12:12.537773+00:00	auto_approval_policy	accept	meta_review	`artifact_9f3842730acf40c7`	yes

A2A/local task ledger

Task	Agent	Stage	State	Output
`task_23f6bc56d927432d`	goal_manager	scope	completed	`artifact_ae850f40e063477b`
`task_487aa0170e0b4b02`	evidence_discovery	generate	completed	`artifact_24d0c17a08104e9a`
`task_863ab811407d449d`	source_verification	generate	completed	`artifact_23c49dd4582546e6`
`task_0d6199a0d6194714`	generation	generate	completed	`artifact_ce9a6e0533bd41ad`
`task_ad50e75c4fe04d54`	reflection	reflect	completed	`artifact_6729074f7f9d49e4`
`task_a405abff8706498a`	novelty_review	reflect	completed	`artifact_9b8d2c50cede41de`
`task_7d0e3411343e4515`	methods_statistics	reflect	completed	`artifact_6abc279774ef4fdc`
`task_6239e7405675409b`	ethics_safety_governance	reflect	completed	`artifact_5cf5fdcceee84d46`
`task_d7e5e316de034b7c`	impact_review	reflect	completed	`artifact_5a95d55e05e54405`
`task_a5d726af4d524ee8`	ranking	rank	completed	`artifact_b05b34f56d8a47f5`
`task_ca95cb4705f845dc`	evolution	evolve	completed	`artifact_afd01fb2c4e04e9d`
`task_ed19c712f3ee41f3`	proximity	proximity	completed	`artifact_623f0b27987f43a4`
`task_436193ff8cce4025`	meta_reviewer	meta_review	completed	`artifact_2bf0c4a3886a4fed`